

Garage Door Project

ME 472 - Mechanical System Design

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Introduction

A garage door is one of the most common systems in the United States and is present in almost every residential home. One of the most common system designs involves a torsion spring, which provides torque to a set of cable drums that use cables to lift the garage door. Attached to the door are rollers and hinges, which assist in its motion. On the tracks are a variety of brackets that hold all the components in place. The objective of this project is to design the mechanical components, conduct stress and fatigue analysis, and factor in safety considerations to develop a reliable and working system that lasts at least 15 years. This includes accounting for typical usage scenarios, load-bearing requirements, kinematic analysis of the opening/closing mechanism, and the integration of components such as torsion springs, hinges, pulleys, and gears. For the system, the door will be 8ft by 8ft and will weigh 100 lbs, which will be segmented into 4 panels. Additionally, it is assumed that the door is opened only twice daily for 15 years. If fully opening and closing are considered one cycle, in 15 years, the total cycles would be 10,950 cycles.

Synthesize and Modeling

Based on stress, deflection, and fatigue analysis, dimensions were determined for the various parts that make up the system. A 3D CAD model of these parts, such as vertical track, horizontal track, and flag bracket, was individually created on SolidWorks. They were assembled using constraints such as coincident and concentric; coincident between points, faces, and edges, and concentric between circular surfaces. For parts designed to move in a certain path, the Path Mate was utilized to create a path for the part to follow. When unexpected gaps occurred in the assembly, temporary plates were used to bridge the gaps

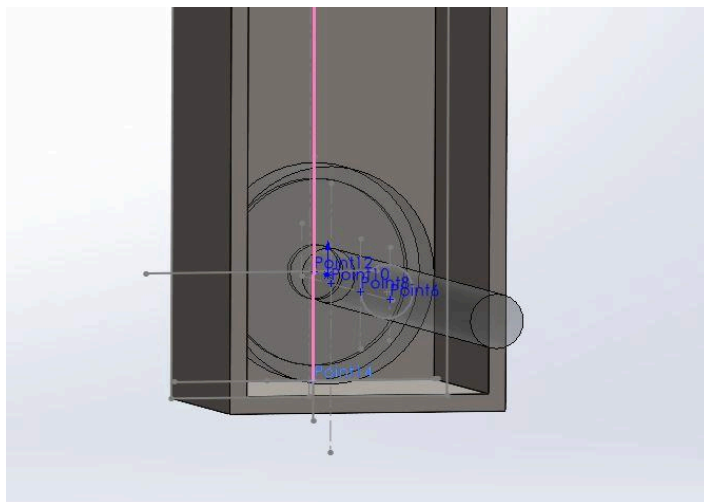


Figure 1 - Path Mate constraint used to ensure that the roller would follow the path sketched in the vertical and horizontal track to serve as the motion of opening and closing the door.

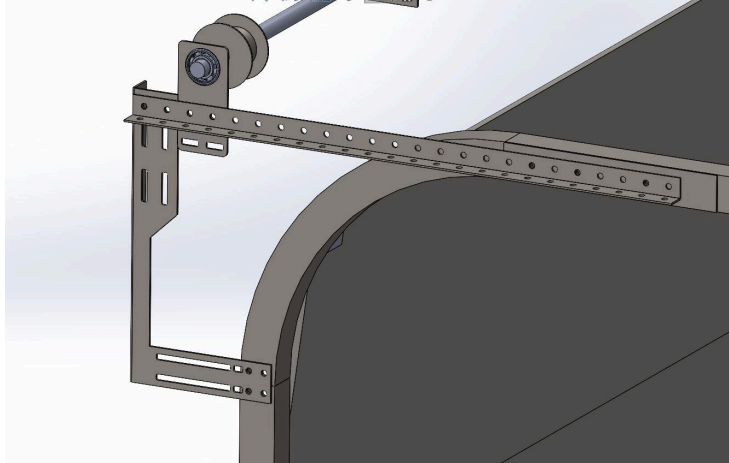


Figure 2 - Coincident and concentric mates used to assemble the flag bracket, angle track, end bearing plate, horizontal, and vertical tracks into place.

Force Analysis

Some of the key forces that act on the system are, firstly, the weight of the door. The door is 8ft by 8ft and weighs 100 lbs. Secondly, there would be the weight of the springs, which, according to an online website, would be approximately 19.83 lb. Then there is the weight of the cable drums, found using the density equation, and the tension in the cable, found through analysis, which results in a force of 48 lbs on each end of the shaft. There is the torque created by the spring, which was selected to be 180 in-lb. Finally, there is the force acting on the rollers, where the sum of that force equals 45 lbs. These forces and torques would be used to determine the stresses, deflection, and fatigue life of each part.

Material Selection

For the brackets, tracks, hinges, and rollers, the material used would be 1045 cold-drawn steel. These components need to be designed to withstand heavy loads and resist deflection. 1045 CD steel offers high strength with a yield strength of 77 kpsi, is wear resistant, and has improved machinability, allowing us to create parts with greater precision. For the spring, the material used was chrome vanadium steel (ASTM A231). This offers great fatigue resistance, high yield strength, and is resistant to permanent deformation under repeated loading. For the cable, the material used to construct it would be galvanized steel grade 37. It has excellent corrosion resistance and is very cost-effective. For the garage door, we used cast alloy steel since it is strong, durable, and has very low maintenance. Finally, for the cable drums, it is constructed out of aluminum 2024 alloy for its high strength, lightweight properties.

Design for Stress

Vertical Track

For the vertical track, it is being constructed out of 1045 CD Steel with a yield stress of 77 kpsi. It is designed as a U-channel to have enough space for the roller to sit and move without obstructions. It is connected to the flag brackets flange with two 0.375 in bolts and will be fixed to the actual door frame itself with 16 bolts spread 1 foot apart from each other. Due to the rollers, we can assume that the vertical track will experience a uniform load of 50 lbs across its length. Using a factor of safety of 2, we assume the allowable stresses to be 38.5 kpsi.

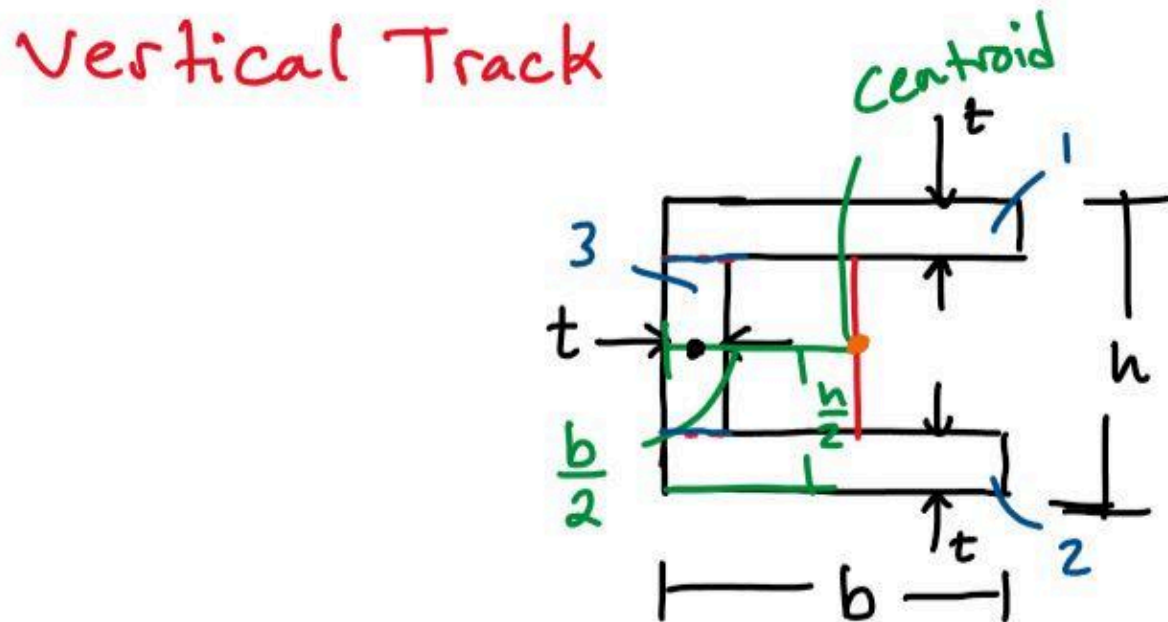


Figure 3 - Top view of vertical track.

This offers a top view of the vertical track and labels its dimensions. To calculate the stresses acting on it, we assumed that the flange base is 1 in, the height is 2 in, and the thickness of the material is 0.109375 in (gauge 12). The length of the vertical track will remain constant at 86 inches.

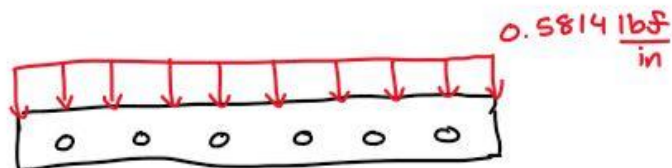


Figure 4 - Load distribution on the vertical track that has been oriented.

The picture above shows how the load is distributed among the vertical tracks. Based on this load, we can see that there is bending stress, bearing stress, and shear stress at the bolts.

$$I = \frac{1}{12}bh^3 + Ad^2$$

$$I_1 = \frac{1}{12}(1\text{in})(0.109375\text{in})^3 + (0.109375\text{in})(0.870625\text{in})^2$$

$$I_1 = I_2 = 0.086867\text{in}^4$$

$$I_3 = \frac{1}{12}bh^3 + \cancel{Ad^2}^{70} = \frac{1}{12}(0.109375\text{in})(1.78125\text{in})^3$$

$$I_3 = 0.0515\text{in}^4$$

$$I_T = I_1 + I_2 + I_3 = 0.2252\text{in}^4$$

$$\omega = \frac{F}{L} = \frac{50\text{lb}_f}{86\text{in}} = 0.5814 \frac{\text{lb}_f}{\text{in}}$$

$$M_{\max} = \frac{\omega L^2}{12} = \frac{(0.5814 \frac{\text{lb}_f}{\text{in}})(86\text{in})^2}{12} = 358\text{lb}_f \cdot \text{in}$$

$$\tau_{\max} = \frac{VQ}{It}$$

$$V_{\max} = 25\text{lb}_f$$

$$Q = A \cdot y' = (1\text{in})(0.109375\text{in})(0.5\text{in}) + (1\text{in})(0.109375\text{in})(0.890625\text{in}) = 0.152\text{in}^3$$

$$I = 0.2252\text{in}^4$$

$$t = 0.109375\text{in}$$

$$\tau_{\max} = \frac{(25\text{lb}_f)(0.152\text{in}^3)}{0.2252\text{in}^4(0.109375\text{in})} = 154.276\text{psi} < 38.5\text{kpsi} \checkmark$$

$$\sigma_{\text{bearing}} = \frac{F}{d \cdot t} = \frac{3.125\text{lb}_f}{(0.375\text{in})(0.109375\text{in})} \quad 16\text{bolts}$$

$$\sigma_{\text{bearing}} = 76.17\text{psi} < 38.5\text{kpsi} \checkmark$$

From these calculations, we can see that the bending, bearing, and shear stress for these dimensions fell within acceptable ranges. Based on these calculations, MATLAB code was written, which enabled the user to input dimensions and would be able to determine whether it

would pass or fail. For our design, we chose a flange base of 2 in, a height of 2 in, and the thickness of the material is 0.140625 in (gauge 10).

Horizontal Track

For the horizontal track, it is being constructed out of the same material as the vertical track. It has the same cross-sectional view as shown in Figure 3 and has a curved section with a radius of 15 in. It is also connected to the flag brackets flange with two 0.375 in bolts as well as the angle track with three 0.375 in bolts. At the end of the track, it connects to the rear track hanger with one 0.375 in bolt, which is fixed to the ceiling. Due to the rollers, we can assume that the horizontal track will experience a uniform load of 50 lbs across its length with the door fully opened. Using a factor of safety of 2, we assume the allowable stresses to be 38.5 kpsi. The load distribution on the horizontal track is the same as shown in Figure 4 when the door is fully opened.

From Figure 4, we can see that the track will also experience bending, shear, and bearing stresses along its length and at its bolts. To calculate the stresses acting on it, we assumed that the flange base is 1 in, the height is 2 in, and the thickness of the material is 0.109375 in (gauge 12). The length of the horizontal track will remain constant at 117 inches.

$$I = 0.2252 \text{ in}^4$$

$$W = \frac{F}{L} = \frac{50 \text{ lbs}}{117 \text{ in}} = 0.427 \frac{\text{lb}}{\text{in}}$$

$$M_{\max} = \frac{WL^2}{12} = \frac{(0.427 \frac{\text{lb}}{\text{in}})(117 \text{ in})^2}{12} = 487 \text{ lb}\cdot\text{in}$$

$$\sigma = \frac{Mc}{I} = \frac{(487 \text{ lb}\cdot\text{in})(1 \text{ in})}{0.2252 \text{ in}^4} = 2163 \text{ psi} < 38.5 \text{ kpsi}$$

$$\tau_{\max} = 154.276 \text{ psi} < 38.5 \text{ kpsi}$$

$$\sigma_{\text{bearing}} = \frac{F}{dt} = \frac{8.33 \text{ lb}}{(0.375 \text{ in})(0.109375 \text{ in})} = 203.17 \text{ psi} < 38.5 \text{ kpsi}$$

By calculating the stresses with the door fully opened, we analyze the track under its maximum conditions and determine the maximum stresses. These calculations show that the moment of inertia and the shear stress are the same. Since they experience the same load and have the same dimensions, the horizontal and vertical tracks would also have the same moment of inertia and shear stress. All three forms of stress fall far short of the acceptable stress values. Similar to the

vertical track, MATLAB code was written, which enabled the user to input dimensions and would be able to determine whether it would pass or fail. For our design, we chose a flange base of 2 in, a height of 2 in, and the thickness of the material is 0.140625 in (gauge 12).

Rear Track Hanger

For the rear track hanger, it is being manufactured out of 1045 CD Steel, which has a yield stress of 77 kpsi. It is designed as an L-bracket, which is then connected to two more L-brackets to provide further support. It is connected to the horizontal track with a 0.375 in bolt and will be fixed to the ceiling or a beam. Using a safety factor of 2, we assume that the allowable stresses are 38.5 kpsi

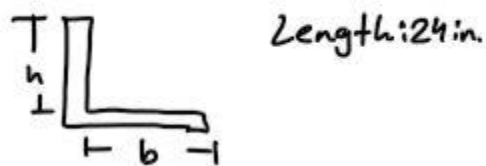


Figure 5 - Side view of rear track hanger.

This offers a side view of the rear track hanger. To calculate the stresses that act on it, we assumed that the flange base was 2 in, the height was 2 in, and the thickness was 0.109375 in. The length of the track will remain constant at 24 inches.

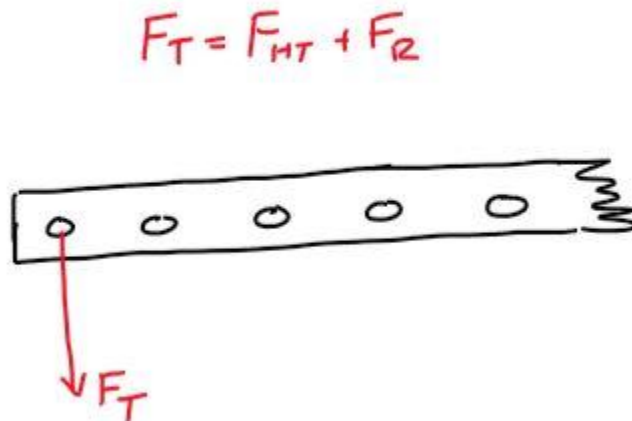
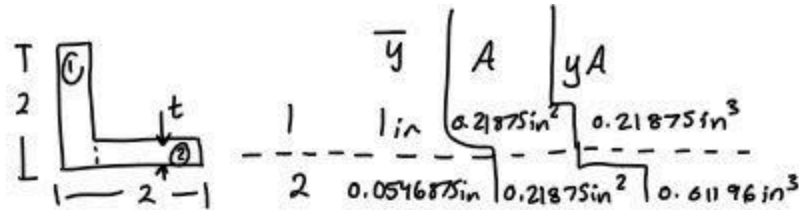


Figure 6- Load distribution on the rear track hanger.

From the load distribution, we can see that a force acts on the rear track hanger. This force is half the weight of the door when fully opened, plus the weight of the supporting bracket. For safety, we assumed the combined force to be 150 lbs, three times the weight of half the weight of the door. We can see that bending stress occurs on the bracket, shear stress and bearing stress occur at the bolts.



$$\Sigma A = 0.4375 \text{ in}^2$$

$$\Sigma YA = 0.2307 \text{ in}^3$$

$$\bar{y} = \frac{\Sigma YA}{\Sigma A} = \frac{0.2307 \text{ in}^3}{0.4375 \text{ in}^2} = 0.527 \text{ in}$$

$$I_1 = \frac{1}{12}bh^3 + Ad^2 = \frac{1}{12}(0.109375 \text{ in})(2 \text{ in})^3 + (0.21875 \text{ in}^2)(0.473 \text{ in})^2$$

$$= 0.1219 \text{ in}^4$$

$$I_2 = \frac{1}{12}bh^3 + Ad^2 = \frac{1}{12}(2 \text{ in})(0.109375 \text{ in})^3 + (0.21875 \text{ in}^2)(0.472 \text{ in})^2$$

$$= 0.04895 \text{ in}^4$$

$$I_T = 0.17085 \text{ in}^4$$

$$M_{\max} = \frac{FL}{8} = \frac{(150 \text{ lbf})(24 \text{ in})}{8} = 450 \text{ lbf} \cdot \text{in}$$

$$\sigma = \frac{Mc}{I} = \frac{(450 \text{ lbf} \cdot \text{in})(1 \text{ in})}{(0.17085 \text{ in}^4)} = 2634 \text{ psi} < 38.5 \text{ kpsi}$$

$$\sigma_{\text{bearing}} = \frac{F}{t \cdot d} = \frac{(150 \text{ lbf})}{(0.109375 \text{ in})(0.375 \text{ in})} = 3657 \text{ psi} < 38.5 \text{ kpsi}$$

$$\tau = \frac{F}{A} = \frac{150 \text{ lbf}}{\frac{\pi}{4}(0.375 \text{ in})^2} = 1358 \text{ psi} < 38.5 \text{ kpsi}$$

By calculating the stresses with the door fully opened, we found the maximum stresses that would act on the rear track hanger. From our calculations, we can see that all the stress values fall below the allowable stress. Like the tracks, MATLAB code was written, which enabled the user to input dimensions and would be able to determine whether it would pass or fail. For our design, we chose a flange base of 1.25 in, a height of 1.25 in, and the thickness of the material is 0.140625 in (gauge 12).

Angle Track

The angle track is being constructed out of 1045 CD steel that has a yield strength of 77 kpsi. It is designed as an L-bracket that is 36 inches long. It connects to the top of the flag bracket with two 0.375 in bolts, the end bearing plate with two 0.375 in bolts, and the horizontal track with three 0.375 in bolts. Using a safety factor of 2, we assume that the allowable stresses are 38.5 kpsi.

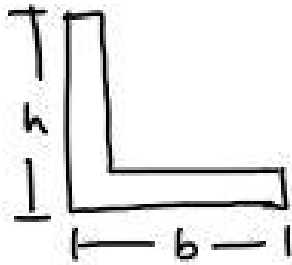


Figure 7 - Side view of rear track hanger.

This provides a side view of the angle track. To calculate the stresses, we assume that the base of the flange is 2 in, the height is 2 in, and the thickness is 0.109375 in. The length will remain constant at 36 inches.

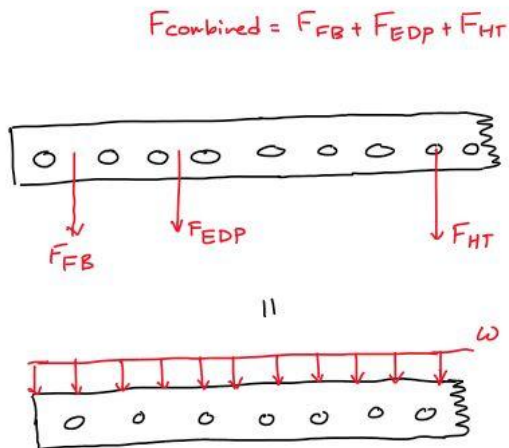


Figure 8 - Load distribution on the angle track.

From the load distribution, we can see that the forces that act on the angle track include the forces from the end bearing plate, the flag bracket, and the horizontal track. We combine the forces to create a net force that we say is acting over the bracket. This will cause bending, bearing, and shear stress on the bracket and the bolts. We say that the net force is 210 lb, the 50 lb from the door, and an assumed force of 20 lb from the end bearing plate and flag bracket, and a factor of safety of 3.

$$I_T = 0.17085 \text{ in}^4$$

$$\omega = \frac{F_{\text{combined}}}{L} = \frac{210 \text{ lbf}}{36 \text{ in}} = 5.83 \frac{\text{lbf}}{\text{in}}$$

$$M_{\text{max}} = \frac{\omega L^2}{12} = \frac{(5.83 \frac{\text{lbf}}{\text{in}})(36 \text{ in})^2}{12}$$

$$= 630 \text{ lbf} \cdot \text{in}$$

$$\sigma = \frac{Mc}{I} = \frac{(630 \text{ lbf} \cdot \text{in})(1 \text{ in})}{0.17085 \text{ in}^4} = 3687 \text{ psi}$$

$$\sigma_b = \frac{F}{d \cdot t} = \frac{90 \text{ lbf}}{(0.375 \text{ in})(0.109375 \text{ in})} = 2194 \text{ psi}$$

$$\tau = \frac{V}{A} = \frac{90 \text{ lbf}}{\frac{\pi}{4}(0.375 \text{ in})^2} = 815 \text{ psi}$$

Our calculations show that the stress values fall far below our allowable stress limits at this dimension. To find working dimensions, MATLAB code was written, which enabled the user to input dimensions and would be able to determine whether it would pass or fail. For our design, we chose a flange base of 1.25 in, a height of 1.25 in, and the thickness of the material is 0.109375 in (gauge 12).

End Bearing Plate

The end bearing plate is being fabricated out of 1045 CD steel that has a yield strength of 77 kpsi. It is designed as a rectangular box with 0.30-inch fillets at the corners. It connects to the angle track with two 0.375 in bolts and has an RLS 8 bearing fitted into a hole near the top center of the plate. Using a safety factor of 2, we assume that the allowable stresses are 38.5 kpsi.

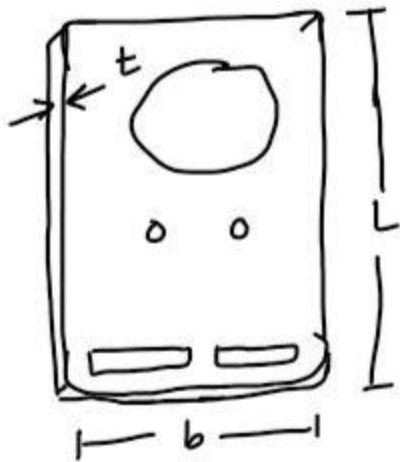


Figure 9 - Top view of end bearing plate

This provides a top view of the end bearing plate. To calculate the stresses, we assume that the base of the plate is 3.25 in, the length is 6.75 inches, and the thickness is 0.140625 in.

$$F = F_D + F_{BW}$$

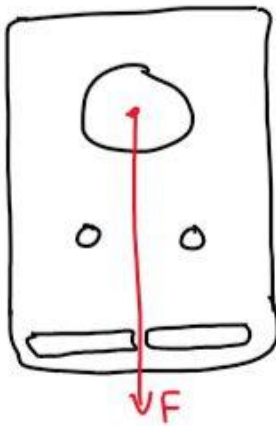


Figure 10- Load distribution on the end bearing plate.

From the load distribution, it becomes apparent that the main driving force causing stresses on the plate is the sum of the radial force and the weight of the bearing. We combine them into a net force and add a factor of safety of 3 to get 174.9 lb. This will cause the plate to experience tension, bearing stresses at the bolts and the bearing hole, and shear stress at the bolts.

$$\sigma = \frac{P}{A} = \frac{174.91b}{(3.25in)(0.140625in)}$$

$$\sigma = 382.7 \text{ psi}$$

$$\sigma_{B_h} = \frac{P}{dt} = \frac{174.91b}{(0.375in)(0.140625in)}$$

$$= 3317 \text{ psi}$$

$$\sigma_{B_B} = \frac{P}{dt} = \frac{174.91b}{(2.25in)(0.140625in)}$$

$$= 552.7 \text{ psi}$$

$$\tau = \frac{V}{A} = \frac{P}{A} = \frac{174.91b}{\frac{\pi}{4}(0.375in)^2}$$

$$= 1584 \text{ psi}$$

The results show that the selected dimension values had stress values that were significantly lower than the allowable stress values.

Shaft Support Bracket

The shaft support bracket is being fabricated out of 1045 CD steel. It is designed as an L-bracket. It connects to the wall using two 0.75-inch bolts. It has a 2.25-inch diameter hole where the bearing is fitted. Using a safety factor of 2, we assume that the allowable stresses are 38.5 kpsi.

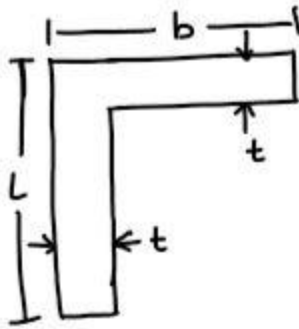


Figure 11 - Top view of shaft support bracket

This provides a top view of the support bracket. To conduct the stress analysis, we assume that the flange base is 2 in, the length is 5.66 in, and the thickness is 0.171875 in.

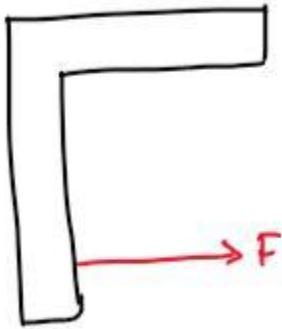
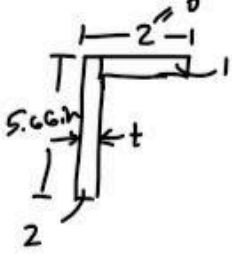


Figure 12 - Load distribution on shaft support bracket.

The load distribution shows us that the main cause behind the stress is the sum of the radial force of the bearing and its weight. We combine them into a net force and add a factor of safety of 3, which leads us to the same force value that was used in the end bearing plate - 174.9 lb. This force causes bending, bearing stress at the bolts and the bearing, and shear stress at the bolts.



$L = 5.66 \text{ in}$
 $b = 2 \text{ in}$
 $t = 0.171875 \text{ in}$

	Y	A	YA
1	5.574 in	0.314 in ²	1.75 in ³
2	2.83 in	0.973 in ²	2.75 in ³
Σ		1.287 in ²	4.5 in ³

$$\bar{Y} = \frac{\Sigma YA}{\Sigma A} = \frac{4.5 \text{ in}^3}{1.287 \text{ in}^2} = 3.50 \text{ in}$$

$$I_1 = \frac{1}{12}bh^3 + Ad^2$$

$$= \frac{1}{12}(1.83 \text{ in})(0.171875 \text{ in})^3 + (0.314 \text{ in}^2)(2.074 \text{ in})^2 = 1.3514 \text{ in}^4$$

$$I_2 = \frac{1}{12}bt^3 + Ad^2$$

$$= \frac{1}{12}(0.171875 \text{ in})(5.66 \text{ in})^3 + (0.973 \text{ in}^2)(0.67 \text{ in})^2 = 3.034 \text{ in}^4$$

$$I_T = 4.3854 \text{ in}^4$$

$$M = F \cdot L = (174.9 \text{ lbf})(2 \text{ in}) = 349.8 \text{ lbf} \cdot \text{in}$$

$$\sigma' = \frac{Mc}{I} = \frac{(349.8 \text{ lbf} \cdot \text{in})(1 \text{ in})}{4.3854 \text{ in}^4} = 264.5 \text{ psi}$$

$$\sigma'_b = \frac{F}{A} = \frac{174.9 \text{ lbf}}{0.375 \text{ in}(0.171875 \text{ in})} = 2713.6 \text{ psi}$$

$$\sigma'_{13.13} = \frac{174.9 \text{ lbf}}{(2.25 \text{ in})(0.171875 \text{ in})} = 452.3 \text{ psi}$$

$$\tau = \frac{F}{A} = \frac{174.9 \text{ lbf}}{\frac{\pi}{4}(0.375 \text{ in})^2} = 1583.6 \text{ psi}$$

The calculation shows that the dimension used resulted in stress values that fell within acceptable values and were significantly lower than the allowable stress values. For our design, we selected a length of 5.66 in, a base of 2 in, and a thickness of 0.171875 in.

Spring

The torsion spring is constructed out of chrome-vanadium which has an elastic modulus of 29.5 MPsi and a shear modulus of 11.2 MPsi. Using available resources, we determined that our spring would have 8.75 turns and an inner diameter of 2 in. The drums available for use in our garage system had 4in, 5.25 in, and 8in diameters. The wire sizes available for the spring ranged from 0.187 in to 0.273 in. Using the door weight and diameter of the cable drums, we determined the torque and used this information to determine the number of coils and spring length. We conducted stress analysis and found the fatigue life and the number of cycles before failure.

Door weight: 100 lbs

I.D.: 2 in

turns: 8.75

Material: Chrome-vanadium $E: 29.5 \text{ MPsi}$, $G = 11.2 \text{ MPsi}$

Drums:

4 in, 5.25 in, 8 in

Wire

0.187 in, 0.192 in, 0.207 in, 0.218 in, 0.225 in,
0.234 in, 0.243 in, 0.250 in, 0.262 in, 0.273 in

$$T = F_{\text{door}} \cdot R_{\text{drum}}$$

$$T_4 = (90 \text{ lbf})(2 \text{ in}) = 180 \text{ lbf} \cdot \text{in}$$

$$T_{5.25} = 236.25 \text{ lbf} \cdot \text{in}$$

$$T_8 = 360 \text{ lbf} \cdot \text{in}$$

$$T = K' \theta \rightarrow K' = \frac{T}{\theta} = \frac{180 \text{ lbf} \cdot \text{in}}{8.75 \text{ turns}} = 20.57 \text{ lb} \cdot \text{in}$$

$$K'_{5.25} = 2716 \cdot \text{in}$$

$$K'_8 = 41.14 \text{ lb} \cdot \text{in}$$

$$K' = \frac{d^4 E}{10.8 D N_a} \rightarrow N_a = \frac{d^4 E}{10.8 D K'}$$

$$d = 0.187 \text{ in}$$

$$D = 2.187 \text{ in}$$

$$N_{a_4} = \frac{(0.187 \text{ in})^4 (29.5 \text{ E6 psi})}{10.8 (2.187 \text{ in}) (20.57 \text{ lb} \cdot \text{in})}$$

$$L_4 = N_{a_4} (d) = (78 \text{ coils}) (0.187 \text{ in}) = 15 \text{ in}$$

$$\approx 78 \text{ coils}$$

$$L_{5.25} = 12 \text{ in}$$

$$L_8 = 8 \text{ in}$$

$$N_{a_{5.25}} \approx 61 \text{ coils}$$

$$N_{a_8} \approx 41 \text{ coils}$$

From our calculations, the stresses placed on the system would cause it to fracture immediately before fatigue can affect it. We wrote MATLAB code to iterate all the wire sizes and cable drum diameters and ultimately choose a spring with a wire size of 0.243 in, 211 total coils, a spring length of 52 in, stress of 139 kpsi, and max torque of 241.18 lbf in. We conducted fatigue analysis on this as well.

$$\sigma_{\max} = 139 \text{ Kpsi} \quad \sigma_{\min} = 20 \text{ Kpsi}$$

$$\sigma_a = \frac{\sigma_{\max} - \sigma_{\min}}{2} = 59.5 \text{ Kpsi}$$

$$S_y = 195 \text{ Kpsi}$$

$$S_{ut} = 224 \text{ Kpsi}$$

$$\sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2} = 79.5 \text{ Kpsi}$$

$$S_e^1 = 100 \text{ Kpsi}$$

$$K_a = (200 \text{ Kpsi})(224 \text{ Kpsi})^{-0.217} = 0.618$$

$$K_b = 0.879(2 \text{ in})^{-0.107} = 0.816$$

$$S_e = K_a K_b S_e^1 = (0.618)(0.816)(100 \text{ Kpsi}) \\ = 50.43 \text{ Kpsi}$$

$$n_f = \left(\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} \right)^{-1} = \left(\frac{59.5 \text{ Kpsi}}{50.43 \text{ Kpsi}} + \frac{79.5 \text{ Kpsi}}{224 \text{ Kpsi}} \right)^{-1}$$

$$n_f = 0.652$$

$$\sigma_{rev} = \frac{59.5 \text{ Kpsi}}{1 - \left(\frac{79.5 \text{ Kpsi}}{224 \text{ Kpsi}} \right)^2} = 68.07 \text{ Kpsi}$$

$$f = 0.77$$

$$a = \frac{(0.77 \cdot 224 \text{ Kpsi})^2}{50.43 \text{ Kpsi}} = 589.9 \text{ Kpsi}$$

$$b = -\frac{1}{3} \log \left(\frac{0.77(224 \text{ Kpsi})}{50.43 \text{ Kpsi}} \right) = -0.178$$

$$N_f = \left(\frac{68.07 \text{ Kpsi}}{589.9 \text{ Kpsi}} \right)^{-\frac{1}{0.178}}$$

$$N_f = 185,657 \text{ cycles}$$

$$(2)(15)(365) = 10950 \text{ cycles}$$

The fatigue analysis shows that the spring will last for 185,637 cycles before failure. This is far above our basic requirement of 10,950 cycles. Therefore, the spring will last for a minimum of 15 years.

Cable Drum

The cable will be constructed out of aluminum 2024, which has an ultimate tensile strength of 42 kpsi and a yield strength of 35 kpsi. For the stress analysis, we assumed the radius of the drum would be 2 inches. Since the door is mechanized, the automated door opener accounts for 10 percent of the weight of the door. Therefore, each cable drum needs to supply a minimum of 45 lbf. That force will create torque as well as a bending moment. Furthermore, since it is under cyclic loading, we will also conduct fatigue analysis on the drum.

$$T = Fr = (45 \text{ lbf})(2 \text{ in}) \\ = 90 \text{ lbf} \cdot \text{in}$$

$$J = \frac{\pi}{2} (2 \text{ in}^4 - 0.5 \text{ in}^4) = 25.03 \text{ in}^4$$

$$\tau_{\max} = \frac{Tc}{J} = \frac{(90 \text{ lbf} \cdot \text{in})(2 \text{ in})}{25.03 \text{ in}^4} = 7.19 \text{ psi}$$

$$M = Fr = (45 \text{ lbf})(2 \text{ in}) = 90 \text{ lbf} \cdot \text{in}$$

$$I = \frac{\pi}{64} (4 \text{ in}^4 - 1 \text{ in}^4) = 12.5 \text{ in}^4$$

$$\sigma = \frac{Mc}{I} = \frac{(90 \text{ lbf} \cdot \text{in})(2 \text{ in})}{(12.5 \text{ in}^4)} = 14.4 \text{ psi}$$

$$\sigma_{\min} = 5 \text{ psi} \quad \sigma_{\max} = 14.4 \text{ psi} \quad \sigma_a = 4.7 \text{ psi}$$

$$S_{ut} = 42 \text{ kpsi}, S_y = 35 \text{ kpsi}$$

$$\sigma'_a = 9.7 \text{ psi}$$

$$S_e' = 21 \text{ kpsi}$$

$$K_a = (200 \text{ kpsi})(42 \text{ kpsi})^{-0.217} = 0.888$$

$$K_b = 0.879(1.48)^{-0.107} = 0.843$$

$$d_e = 0.370(4) \\ = 1.48$$

$$S_e = K_a K_b S_e' \\ = (0.888)(0.843)(21 \text{ kpsi}) \\ = 15.7 \text{ kpsi}$$

$$n_f = \left(\frac{4.7 \text{ psi}}{15,700 \text{ psi}} + \frac{9.7 \text{ psi}}{42,000 \text{ psi}} \right)^{-1} = 1886 \rightarrow \text{infinite life}$$

Our calculation showed that our stress values using the 4-inch diameter drum were far below the yield strength. Fatigue analysis determined that the part has infinite life. For our design, we ultimately decided to use the 4-inch diameter cable drum.

Cable

The steel cable will be constructed out of galvanized steel grade 37, which has a yield strength of 37 kpsi and an ultimate tensile strength of 52 kpsi. We use the cable drum's radius and the door weight to find the tension needed on the cable. We reverse engineer this for the cable's diameter. We also conduct fatigue analysis on the cable since it experiences cyclic load, and it becomes necessary to determine whether it will fail under fatigue.

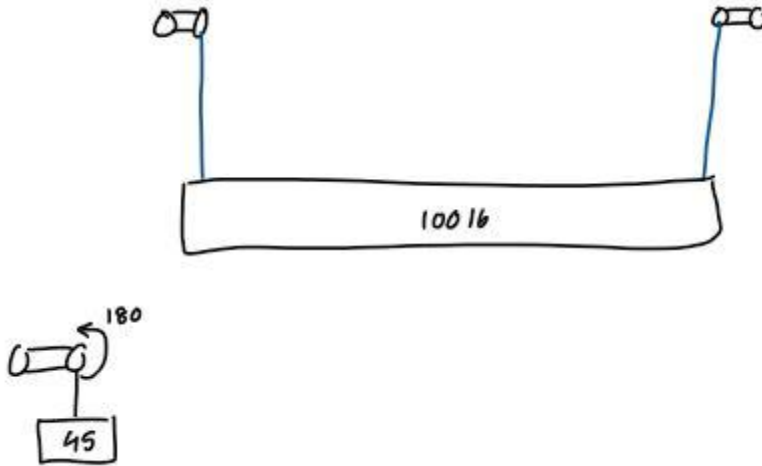


Figure 13 - Free body diagram for the cable drum.

$$\begin{aligned}
 T &= F \cdot r \\
 &= (45 \text{ lb})(2 \text{ in}) \\
 &= (90 \text{ in-lb}) \text{ per drum}
 \end{aligned}$$

Material: Galvanized Steel Grade 37

$$S_y = 37 \text{ kpsi}$$

$$S_{ut} = 52 \text{ kpsi}$$

$$\sigma_{\text{allowable}} = \frac{S_{ut}}{n} = \frac{52 \text{ kpsi}}{3} = 17.33 \text{ kpsi}$$

$$\sigma_{\text{allowable}} = \frac{F}{A} \rightarrow A = \frac{F}{\sigma_{\text{allowable}}} = \frac{(45 \text{ lb})}{(17.33 \text{ kpsi})} = 0.00260 \text{ in}^2$$

$$\frac{\pi d^2}{4} = 0.00260 \text{ in}^2$$

$$d = 0.0575 \text{ in}$$

$$= \frac{1}{16} \text{ in}$$

$$\sigma'_{\min} = 5.3 \text{ psi} \quad \sigma'_{\max} = \frac{Mc}{I} = \frac{(90 \text{ in-lb})(2 \text{ in})}{\frac{\pi}{64}(4 \text{ in})^4} = 14.3 \text{ psi}$$

$$S_e = 26 \text{ kpsi} \quad \sigma'_a = 4.5 \text{ psi}$$

$$K_a = (2.00 \text{ kpsi})(52 \text{ kpsi})^{-0.217} = 0.849 \quad \sigma'_m = 9.8 \text{ psi}$$

$$K_b = 1.315$$

$$d_e = 0.370\left(\frac{1}{16}\right) = 0.023125$$

$$\begin{aligned}
 S_e &= K_a K_b S_e = (0.849)(1.315)(26 \text{ kpsi}) \\
 &= 29 \text{ kpsi}
 \end{aligned}$$

$$n = \left(\frac{4.5 \text{ psi}}{29,000 \text{ psi}} + \frac{9.8 \text{ psi}}{52,000 \text{ psi}} \right)^{-1} = 2910 \rightarrow \text{infinite life}$$

From the calculation, we can see that the diameter needed to satisfy the allowable stress is about 0.0625 inches. This is still significantly less than the yield strength of the material, and the fatigue analysis further confirms this finding by showing that these stress values would lead to infinite life. For our design, we settled on using the 0.0625-inch diameter cable.

Shaft

The solid shaft was fabricated out of 1045 CD steel. It has a length of 100.5 inches and has two support brackets placed - one 78.25 inches away from the left side and a second one placed 22.25 inches from the left. It has a total weight of 115.83 lb acting on the shaft; 45 lb of tension, 3 lb of weight from the cable drums, and 19.86 lb from the spring. We want the deflection of the shaft to be less than 0.1 inch and the angle of twist to be less than 1 degree.



$$y_{\max} \leq 0.1 \text{ in} \quad \phi \leq 1^\circ$$

$$I = \frac{\pi (1 \text{ in})^4}{64} = 0.04909 \text{ in}^4$$

$$y_{\max} = \frac{FL^3}{192EI} = \frac{(115.83 \text{ lb})(54 \text{ in})^3}{192(30 \times 10^6 \text{ psi})(0.04909 \text{ in}^4)}$$

$$y_{\max} = 0.06 \text{ in} \leq 0.1 \text{ in}$$

$$\theta = 1^\circ = 0.01745 \text{ radians}$$

$$\theta = \frac{TL}{JG} \rightarrow J = \frac{TL}{\theta G} = \frac{(180 \text{ in lb})(100.5 \text{ in})}{(0.01745)(11.5 \times 10^6 \text{ psi})}$$

$$J = 0.090 \text{ in}^4$$

$$\frac{\pi d^4}{32} = 0.090 \text{ in}^4$$

$$d = 0.98 \text{ in} \approx 1 \text{ in}$$

We found the deflection between the longest unsupported length, which occurred between the two support brackets. Our deflection was less than 0.1 in, and in order for the angle of twist to be less than 1 degree, the minimum diameter shaft has to be 0.98. We assumed the diameter of the shaft to be 1 in and began stress and fatigue analysis.

$$\begin{aligned}
 M_{max} &= \frac{FL}{4} = \frac{(48 \text{ lbf})(100 \text{ Sin})}{4} = 1206 \text{ lbf} \cdot \text{in} & \tau &= \frac{T_c}{J} & J &= \frac{\pi}{32} (1 \text{ in})^4 \\
 I &= \frac{\pi d^4}{64} = \frac{\pi (1 \text{ in})^4}{64} = 0.04909 \text{ in}^4 & \tau &= \frac{(180 \text{ in} \cdot \text{lbf})(0.5 \text{ in})}{0.04909 \text{ in}^4} & J &= 0.09817 \text{ in}^4 \\
 & & & & T &= 180 \text{ in} \cdot \text{lbf} \\
 \sigma &= \frac{Mc}{I} = \frac{(1206 \text{ lbf} \cdot \text{in})(0.5 \text{ in})}{0.04909 \text{ in}^4} \approx 12,284 \text{ psi} & \tau &= 717 \text{ psi} \\
 \sigma_{min} &= 5 \text{ Kpsi} & \sigma_{max} &= 12,284 \text{ Kpsi} \\
 \sigma'_m &= 8.642 \text{ Kpsi} & \sigma'_a &= 3.642 \text{ Kpsi} & S_{ut} &= 91 \text{ Kpsi} \\
 S_e' &= 45.5 \text{ Kpsi} & & & S_y &= 77 \text{ Kpsi} \\
 K_a &= (2.00 \text{ Kpsi})^{-0.217} = 0.751 \\
 K_b &= 0.879(0.370)^{-0.107} = 0.978 \\
 c_e &= 0.370 \\
 S_e &= S_e' K_a K_b = (45.5 \text{ Kpsi})(0.751)(0.978) \\
 S_e &= 33.4 \text{ Kpsi} \\
 n_f &= \left(\frac{3.642 \text{ Kpsi}}{33.4 \text{ Kpsi}} + \frac{8.642 \text{ Kpsi}}{33.4 \text{ Kpsi}} \right)^{-1} \\
 &= 2.72 \rightarrow \text{infinite life}
 \end{aligned}$$

The results of the analysis show that the stresses for a 1-inch diameter shaft is less than the allowable stress values. Furthermore, the fatigue analysis found that the part would achieve infinite life. For our design, we ultimately decided to use a 1-inch diameter solid shaft.

Bearings

For the bearings, we used the SKF catalog to determine which bearing best suited our situation. We calculated the radial load acting on the bearing and the desired revolutions of our bearing. This allowed us to find the dynamic load rating, which we then used to find the bearing. We used a different equation to find the dynamic load rating and received a similar result. We also conducted fatigue analysis to determine how many revolutions and hours the bearing would last.

$$\begin{aligned}
 R_D &= \frac{F_{\text{spring}}}{2} + (F_{\text{orum}} + F_{\text{tension}}) \\
 &= \frac{19.831 \text{ lbf}}{2} + (481 \text{ lbf}) \\
 &= 57.9216 (2) = 115.8316
 \end{aligned}$$

$$\text{SKF} \rightarrow L_R = 10^6 \text{ revolutions}$$

$$L_D = (365)(15)(24) = 10950 \text{ hrs}$$

$$V_D = 7 \frac{\text{in}}{\text{s}} \quad \omega = \frac{V}{r} = \frac{7 \text{ in/s}}{2 \text{ in}} = 3.5 \text{ rad/s}$$

$$n_D = \omega \left(\frac{60}{2\pi} \right) = 33.42 \frac{\text{rev}}{\text{min}}$$

$$C_{10} = R_D \left[\frac{L_D n_D 60}{L_R} \right]^{\frac{1}{a}} = (115.8316) \left[\frac{(10,950 \text{ hrs})(33.42 \frac{\text{rev}}{\text{min}})(60 \text{ min})}{1,000,000 \text{ rev}} \right]^{\frac{1}{3}}$$

$$C_{10} = 32416 \text{ lbf}$$

$$\begin{aligned}
 C_{10} &= a_f R_D \left[\frac{x_0}{x_0 + (\theta - x_0)(1 - R_0)^{1/b}} \right]^{\frac{1}{a}} \\
 C_{10} &= 115.8316 \text{ lbf} \left[\frac{21.957}{(0.02) + (4.459 - 0.02)(1 - 0.95)^{1/483}} \right]^{\frac{1}{3}} \\
 C_{10} &= 379
 \end{aligned}$$

$a_f = 1$
 $x_D = 21.95694$
 $x_0 = 0.02$
 $\theta = 4.459$
 $a = 3$
 $b = 1483$
 $R_0 = 0.95$

$$L_{10} = \left(\frac{C}{P} \right)^P = \left(\frac{40021 \text{ lbf}}{115.8316 \text{ lbf}} \right)^3$$

$$L_{10} = 41,245 \text{ (millions of rev)}$$

$$L_{10h} = \frac{10^6}{10(33.42 \frac{\text{rev}}{\text{min}})} [41,245] = 123,414,125 \text{ hours}$$

From the dynamic load rating, we selected the RLS 8 deep groove ball bearing for our design, which has a rating of 4002 lbf, almost 10 times the value we determined. Furthermore, the bearing is rated to last for over 100 million hours, far exceeding our desired value of 10,950 hours.

Roller

The rollers are designed to help guide the garage door along the tracks and help open and close the door. The rollers will be made out of 1045 CD steel. It will have a shaft length of 5.6 in. For safety purposes, each roller will be designed to handle half the weight of the door, 50 lbs. To determine the maximum possible stress that can occur on the roller, we will model it as a cantilever beam. This will allow us to design the roller to ensure it never fractures or fails. Under the load of the door, the roller will experience bending, shear, and bearing stress. Additionally, we will also complete fatigue analysis since it experiences cyclic load, and we need to ensure it doesn't fail under fatigue. To conduct the analysis, we assumed that the roller had a diameter of 1.75 in, a thickness of 0.5 in, and the shaft had a diameter of 0.4375 in. The shaft length remains constant.

$$\tau = \frac{F}{2A} = \frac{50 \text{ lb}}{2 \left(\frac{\pi}{4} (0.4375 \text{ in})^2 \right)} = 166.3 \text{ psi}$$

$$\sigma_{\text{Bending}} = \frac{Mc}{I} = \frac{(50 \text{ lb})(5.6 \text{ in})(0.21875 \text{ in})}{\frac{\pi}{64} (0.4375 \text{ in})^4}$$

$$\sigma_{\text{Bending}} = 34058 \text{ psi} \approx 34.058 \text{ Kpsi}$$

$$\sigma_{\text{bearing}} = \frac{F}{d \cdot t} = \frac{50 \text{ lb}}{(0.4375 \text{ in})(0.140625 \text{ in})} = 812.7 \text{ psi}$$

$$\sigma_{\text{min}} = 5 \text{ Kpsi} \quad \sigma_{\text{max}} = 34.058 \text{ Kpsi} \quad S_{\text{ut}} = 91 \text{ Kpsi}$$

$$\sigma_m = 19.529 \text{ Kpsi}$$

$$\sigma_a = 14.529 \text{ Kpsi}$$

$$S_y = 77 \text{ Kpsi}$$

$$S_e^1 = 49.5 \text{ Kpsi}$$

$$K_a = (200 \text{ Kpsi})(91 \text{ Kpsi})^{-0.217} = 0.751$$

$$K_b = 0.879(0.161875 \text{ in})^{-0.107} = 1.068$$

$$d_e = 0.161875 \text{ in}$$

$$\begin{aligned} S_e &= K_a K_b S_e^1 \\ &= (0.751)(1.068)(49.5 \text{ Kpsi}) \\ &= 39.7 \text{ Kpsi} \end{aligned}$$

$$\begin{aligned} n_s &= \left(\frac{14.529 \text{ Kpsi}}{39.7 \text{ Kpsi}} + \frac{19.529 \text{ Kpsi}}{91 \text{ Kpsi}} \right)^{-1} \\ &= 1.72 \rightarrow \text{infinite life} \end{aligned}$$

The calculations show that with these dimensions, the stress values fall below the allowable stress values, allowing the part to function as intended without fracturing. For our design, we

ultimately decided to use a diameter of 1.75 in, a thickness of 0.5 in, and a shaft with a diameter of 0.4375 in.

Hinges

The hinges are designed to allow the garage door to bend when moving along the track. It will be manufactured out of 1045 CD steel and will be connected using a 0.4375-inch bolt. Both plates will be bolted to the garage door using 0.75 in bolts. The top hinge plate is slightly wider than the bottom hinge plate, but both must have the same length. The roller will fit through the hole in the top plate. The rollers for our system were designed to handle half the weight of the door, and this load was transferred to the hinges. As a result, the load acting on the hinge is 50 lbs. This load causes bearing stress between the two hinges, between the roller and the top plate, between the hinges and the garage door, bending stress at the top plate, and shear stress at the top plate. We also conducted fatigue analysis on the hinges.

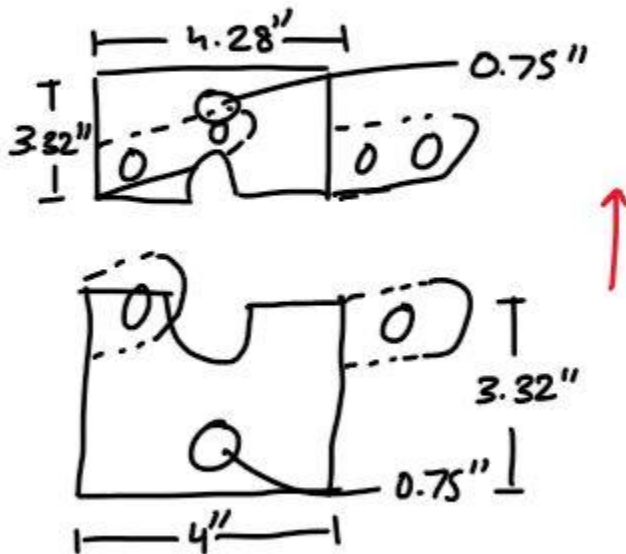


Figure 14 - Dimensions of the hinges, thickness is 0.140625 in.

$$\sigma_{\text{Bearing}} = \frac{F}{d \cdot t} = \frac{25 \text{ lbf}}{(0.4375 \text{ in})(0.140625 \text{ in})} = 406.3 \text{ psi}$$

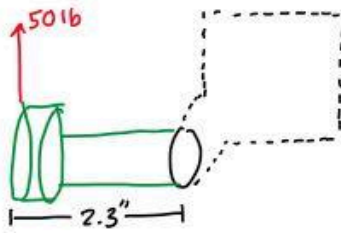
$$\sigma_{\text{Bearing b/w roller and top plate}} = 812.7 \text{ psi}$$

$$\sigma_{\text{Bearing Hinge}} = \frac{F}{d \cdot t} = \frac{25 \text{ lb}}{(0.75 \text{ in})(0.140625 \text{ in})} = 237.04 \text{ psi}$$

$$\sigma_{\text{Bearing Door}} = \frac{F}{d \cdot t} = \frac{25 \text{ lb}}{(0.75 \text{ in})(0.5 \text{ in})} = 66.7 \text{ psi}$$

$$\tau = \frac{V}{2A} = \frac{F}{2\left(\frac{\pi}{4} D^2\right)} = 166.3 \text{ psi}$$

For the bearing stress between the two plates, we assumed it was an ideal system so that the 50 lb load was shared evenly between the top and bottom plates. Therefore, when calculating the bearing stress, we used 25 lb instead of 50 lb.



$$M = F \cdot L = (50 \text{ lb})(2.3 \text{ in}) = 115 \text{ lb}\cdot\text{in}$$

$$I = \frac{1}{12} b t^3 = \frac{1}{12} (4.28 \text{ in})(0.140625 \text{ in})^3$$

$$= 9.919 \times 10^{-4} \text{ in}^4$$

$$\sigma_{\text{bending}} = \frac{M c}{I} = \frac{(115 \text{ lb}\cdot\text{in})(0.0703125 \text{ in})}{(9.919 \times 10^{-4} \text{ in}^4)}$$

$$= 8152.3 \text{ psi}$$

$$\sigma_{\text{max}} = 8.152 \text{ kpsi} \quad \sigma_{\text{min}} = 5 \text{ kpsi}$$

$$\sigma_m = 6.576 \text{ kpsi}$$

$$\sigma_a = 1.576 \text{ kpsi}$$

$$S_e' = 45.5 \text{ kpsi}$$

$$K_a = (2.00 \text{ kpsi})(0.1 \text{ kpsi})^{-0.217} = 0.751$$

$$K_b = 0.91(3.05)^{-0.157} = 0.764$$

$$d_e = 0.808 (w_b)^{\frac{1}{2}} = 3.05$$

$$S_e = K_a \cdot K_b \cdot S_e'$$

$$= (0.751)(0.764)(45.5 \text{ kpsi})$$

$$= 26.1 \text{ kpsi}$$

$$n_f = \left(\frac{1.576 \text{ kpsi}}{26.1 \text{ kpsi}} + \frac{6.576 \text{ kpsi}}{91 \text{ kpsi}} \right)^{-1}$$

$$= 7.54$$

From our calculations, we can see that with the dimensions used, the stress values were less than the allowable stress values. These dimensions will allow the part to function as intended without fracturing and failing. Furthermore, the fatigue analysis indicates infinite life, further supporting the idea. For our design, the top plate had a length of 4.28 inches and a height of 3.32 inches. The bottom plate had a length of 4 inches and a height of 3.32 inches. They both have a thickness of 0.140625 inches.

Roller Bracket

The roller bracket was designed to be attached to the bottom of the first panel and the top of the last panel. Since these panels would not have hinges attached to them, a bracket was needed in order for the rollers to have somewhere to sit. The roller bracket is attached to the garage door with a 0.75-inch bolt. It experiences a load of 50 lb from the roller, which causes it to experience bending, bearing, and shear stress.

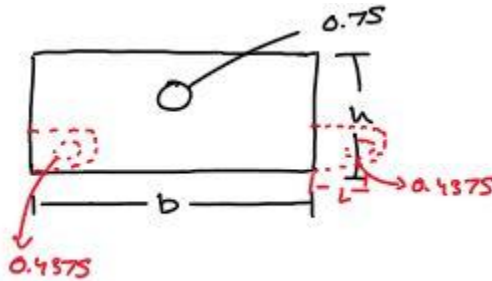


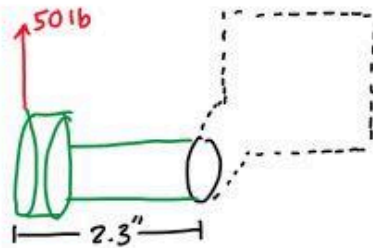
Figure 15 - Front view of the roller bracket

In order to conduct stress analysis, we assume that the bracket has a base of 4 inches, a height of 2.32 inches, and a flange length of 0.38 inches.

$$\sigma = \frac{F}{A} = \frac{50 \text{ lbf}}{(0.4375)(0.140625 \text{ in})} = 812.7 \text{ psi}$$

$$\sigma_{\text{bearing @ Dashed}} = \frac{F}{A} = \frac{50 \text{ lbf}}{(0.75 \text{ in})(0.140625 \text{ in})} = 474 \text{ psi}$$

$$\tau = \frac{F}{2A} = \frac{50 \text{ lbf}}{2\left(\frac{\pi}{4}(0.4375 \text{ in})^2\right)} = 166.3 \text{ psi}$$



$$M = F \cdot L = (50 \text{ lbf})(2.3 \text{ in}) = 115 \text{ lbf} \cdot \text{in}$$

$$I = \frac{1}{12} b t^3 = \frac{1}{12} (4 \text{ in})(0.140625 \text{ in})^3 = 9.27 \text{ E-4 in}^4$$

$$\sigma = \frac{M c}{I} = \frac{(115 \text{ lbf} \cdot \text{in})(0.0703125 \text{ in})}{9.27 \text{ E-4 in}^4} = 8723 \text{ psi} = 8.72 \text{ Kpsi}$$

Since the roller bracket also experiences cyclic loading, we need to conduct fatigue analysis in order to determine if the part fails under fatigue.

$$\sigma_{\max} = 8.72 \text{ kpsi} \quad \sigma_{\min} = 5 \text{ kpsi}$$

$$\sigma_m = 6.86 \text{ kpsi}$$

$$\sigma_a = 1.86 \text{ kpsi}$$

$$S_e' = 45.5 \text{ kpsi}$$

$$K_a = (2.00 \text{ kpsi})(911 \text{ kpsi})^{-0.217} = 0.751$$

$$K_b = 0.91(2.46)^{-0.157} = 0.79$$

$$d_e = 2.46$$

$$S_e = K_a K_b S_e' = (0.79)(0.751)(45.5 \text{ kpsi})$$

$$= 26.99 \text{ kpsi}$$

$$n_f = \left(\frac{1.86 \text{ kpsi}}{26.99 \text{ kpsi}} + \frac{6.86 \text{ kpsi}}{91 \text{ kpsi}} \right)^{-1}$$

$$= 6.93$$

The calculations demonstrate that with these dimensions, the stress values are less than the allowable stress and fall within acceptable ranges. The fatigue analysis indicates infinite life, allowing us to use these dimensions to design the part. In our case, we used a base of 4 inches, a height of 2.32 inches, and a flange length of 0.38 inches to design our part.

Flag Bracket

The flag bracket was designed to connect the horizontal and vertical components using its flange and will be constructed out of 1045 CD steel. It also serves to anchor the end bearing plate and angle track since it is fixed to a wall. It's connected to the horizontal track with two 0.375 in bolts, the vertical track with two 0.375 in bolts, and the angle track with two 0.375 in bolts. Furthermore, it is fixed to the wall using bolts that fit into the slot holes of the flag bracket.

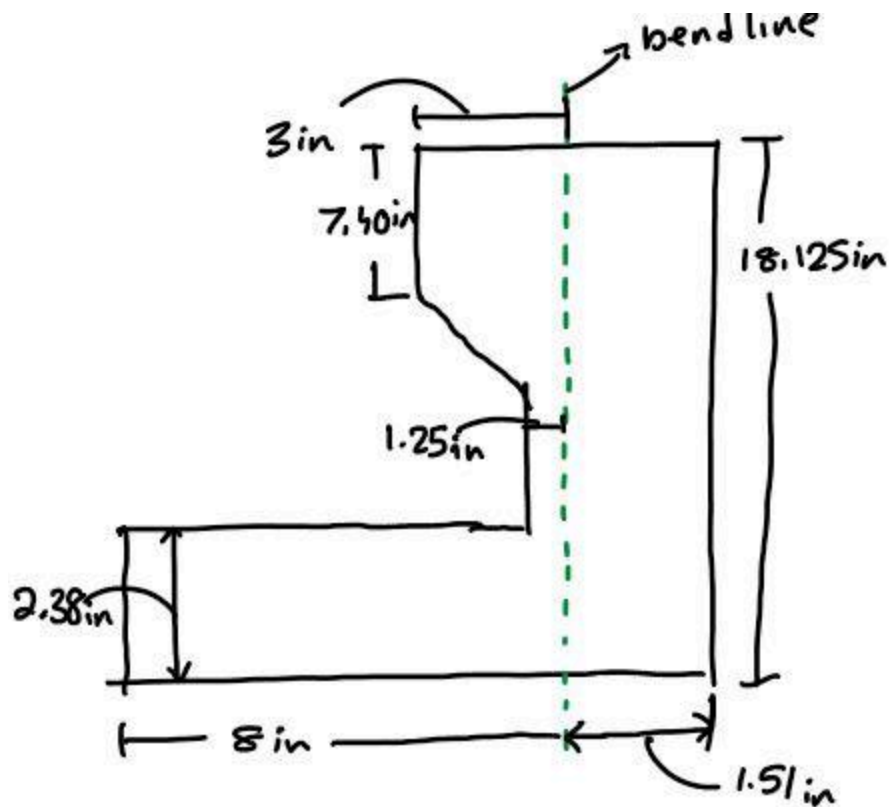


Figure 16 - The dimensions of the flag bracket

The load acts at the bolt holes on the flange. The magnitude of that load changes depending on where the garage door is located. Therefore, in addition to conducting stress analysis, we also need to do fatigue analysis.

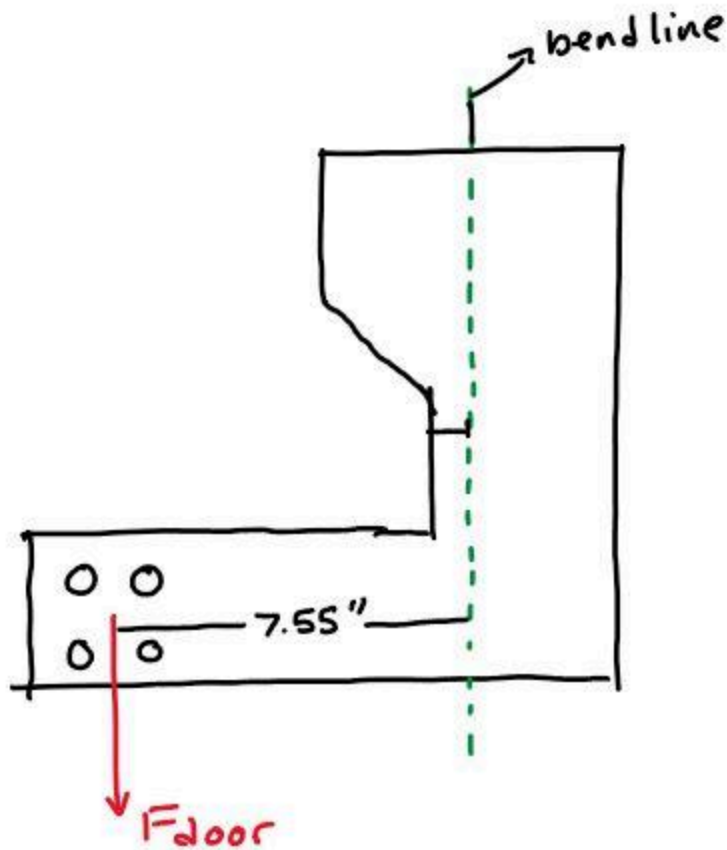


Figure 17 - The free body diagram of the flag bracket

We assume that the force exerted by the garage door is the maximum of 50 lbs. Therefore, based on the free body diagram, we can determine that the bracket experiences bending stress, bearing stress, and shear stress at the bolts.

$$\sigma_{\text{Bearing}} = \frac{F}{d \cdot t} = \frac{50 \text{ lbf}}{(0.375 \text{ in})(0.140625 \text{ in})} = 9481 \text{ psi}$$

$$M = F \cdot L = (50 \text{ lbf})(7.5 \text{ in}) = 375 \text{ lbf} \cdot \text{in}$$

$$I = \frac{1}{12} b t^3 = \frac{1}{12} (8 \text{ in})(0.140625 \text{ in})^3 = 0.00185 \text{ in}^4$$

$$\sigma = \frac{Mc}{I} = \frac{(375 \text{ lbf} \cdot \text{in})(0.0703125 \text{ in})}{0.00185 \text{ in}^4} = 14,317 \text{ psi}$$

$$\tau = \frac{V}{A} = \frac{50 \text{ lbf}}{\frac{\pi}{4} (0.5 \text{ in})^2} = 452.7 \text{ psi}$$

$$\sigma_{\text{max}} = 14.317 \text{ kpsi} \quad \sigma_{\text{min}} = 5 \text{ kpsi}$$

$$\sigma_m = 9.6585 \text{ kpsi}$$

$$\sigma_a = 4.6585 \text{ kpsi}$$

$$S_e' = 45.5 \text{ kpsi}$$

$$K_a = (2.00 \text{ kpsi})(91 \text{ kpsi})^{-0.217} = 0.751$$

$$K_b = 0.91(3.52)^{-0.157} = 0.747$$

$$d_e = 3.52 \text{ in}$$

$$S_e = K_a K_b S_e' = (0.751)(0.747)(45.5 \text{ kpsi}) = 25.5 \text{ kpsi}$$

$$n_f = \left(\frac{4.6585 \text{ kpsi}}{25.5 \text{ kpsi}} + \frac{9.6585 \text{ kpsi}}{91 \text{ kpsi}} \right)^{-1}$$

$$n_f = 3.46 \rightarrow \text{infinite life.}$$

The results show that stress values fall within acceptable ranges and are below the allowable stress values for this material. The fatigue analysis shows the material will be able to function as intended without experiencing fatigue failure. For our design, we utilized the same dimensions shown in Figure 16.

Design for Deflection

The vertical, horizontal tracks, rear track hanger, and angle track are all affected by deflection. Therefore, it was crucial that we designed the parts to ensure that deflection did not exceed 0.1 inches. If it began to exceed this value, it would begin to interfere with the system's operation, decreasing reliability and the life of the system as a whole.

Vertical Track

$$y_{\max} \leq 0.1 \text{ in}$$

$$y_{\max} = \frac{wL^4}{384EI} = \frac{(0.5814 \frac{\text{lb}}{\text{in}})(86 \text{ in})^4}{384(30 \text{ E}6 \text{ psi})(0.2252 \text{ in}^4)}$$

$$y_{\max} = 0.012 \text{ in} \checkmark$$

With a flange base of 1 in, a height of 2 in, and a thickness of 0.109375, the vertical track was able to stay below the deflection limit. MATLAB code was created that allows users to insert combinations of dimensions, and the code will determine if it passes the stress and deflection limits.

Horizontal Track

$$y_{\max} \leq 0.1$$

$$y_{\max} = \frac{wL^4}{384EI} = \frac{(0.427 \frac{\text{lb}}{\text{in}})(117 \text{ in})^4}{384(30 \text{ E}6 \text{ psi})(0.2252 \text{ in}^4)}$$

$$y_{\max} = 0.03 \text{ in}$$

With dimensions identical to the vertical track, the horizontal track also manages to stay below the deflection limit. Like the vertical track, the horizontal track also has MATLAB code which allows the user to upload dimensions and it determines whether it passes or fails.

Rear Track Hanger

$$\gamma_{\max} = \frac{FL^3}{192EI} = \frac{(150\text{lbF})(24\text{in})^3}{192(30\text{E}6)(0.1708\text{in}^4)}$$

$$\gamma_{\max} = 0.002\text{ in}$$

An L-shaped bracket with a base of 2 in, a height of 2in, and a thickness of 0.109375in, it too manages to stay below the deflection limit. It also has MATLAB code.

Angle Track

$$\gamma_{\max} = \frac{\omega L^4}{384EI} = \frac{(5.83\frac{\text{lbF}}{\text{in}})(36\text{in})^4}{384(30\text{E}6\text{psi})(0.1708\text{in}^4)}$$

$$\gamma_{\max} = 0.005\text{ in}$$

An L-shaped bracket with identical dimensions to the rear track hanger it too manages to stay below the deflection limit. It also has MATLAB code.

Conclusion

This project successfully designed an effective and reliable garage door system that will operate safely for the next 15 years. Through detailed stress, fatigue, and deflection analysis, each part was thoroughly inspected and designed to withstand standard loading conditions while maintaining safety factors. Material selection prioritized strength, fatigue resistance, and cost-effectiveness. Certain results were validated using MATLAB code and simulations. The parts were modeled in SolidWorks with the proper constraints to replicate realistic operation. This garage door system will provide reliable operation for the next 15 years.

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Appendix

A. Technical Drawings

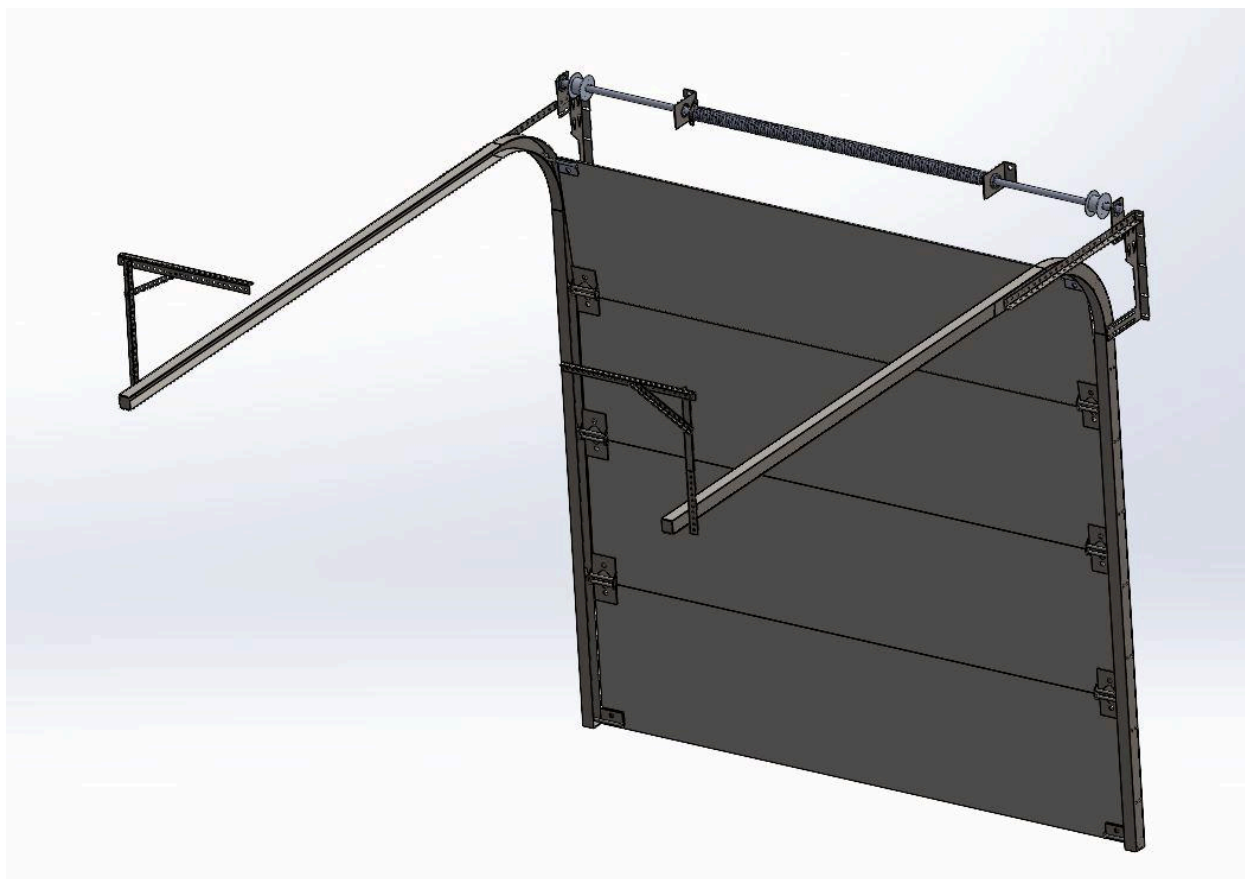


Figure 18 - Fully assembled garage door

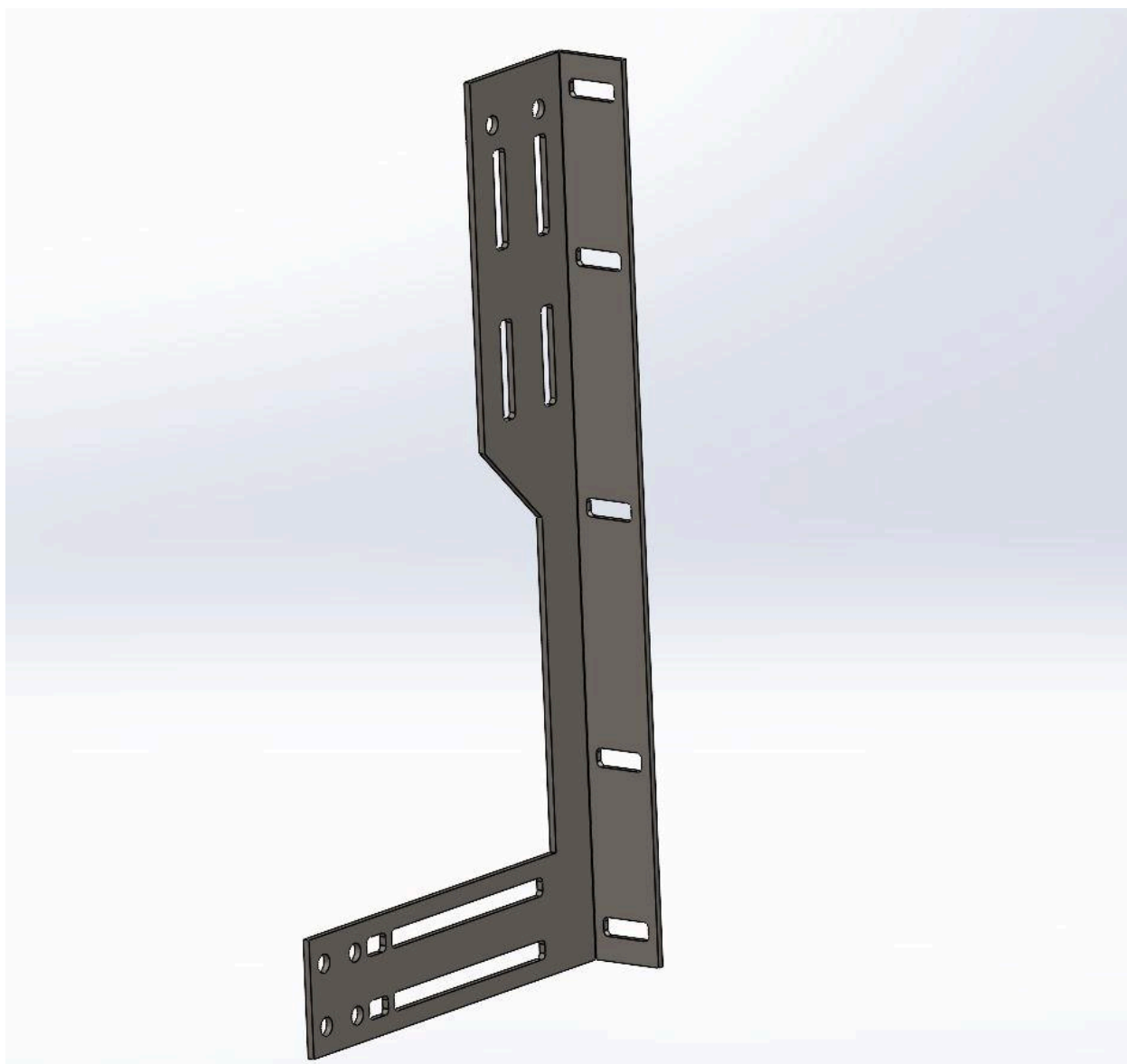


Figure 19 - Flag bracket

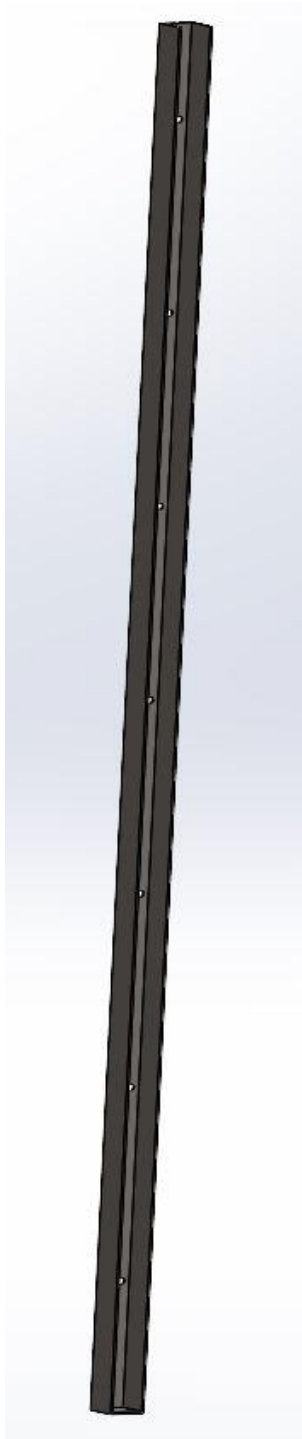


Figure 20 - Vertical Track

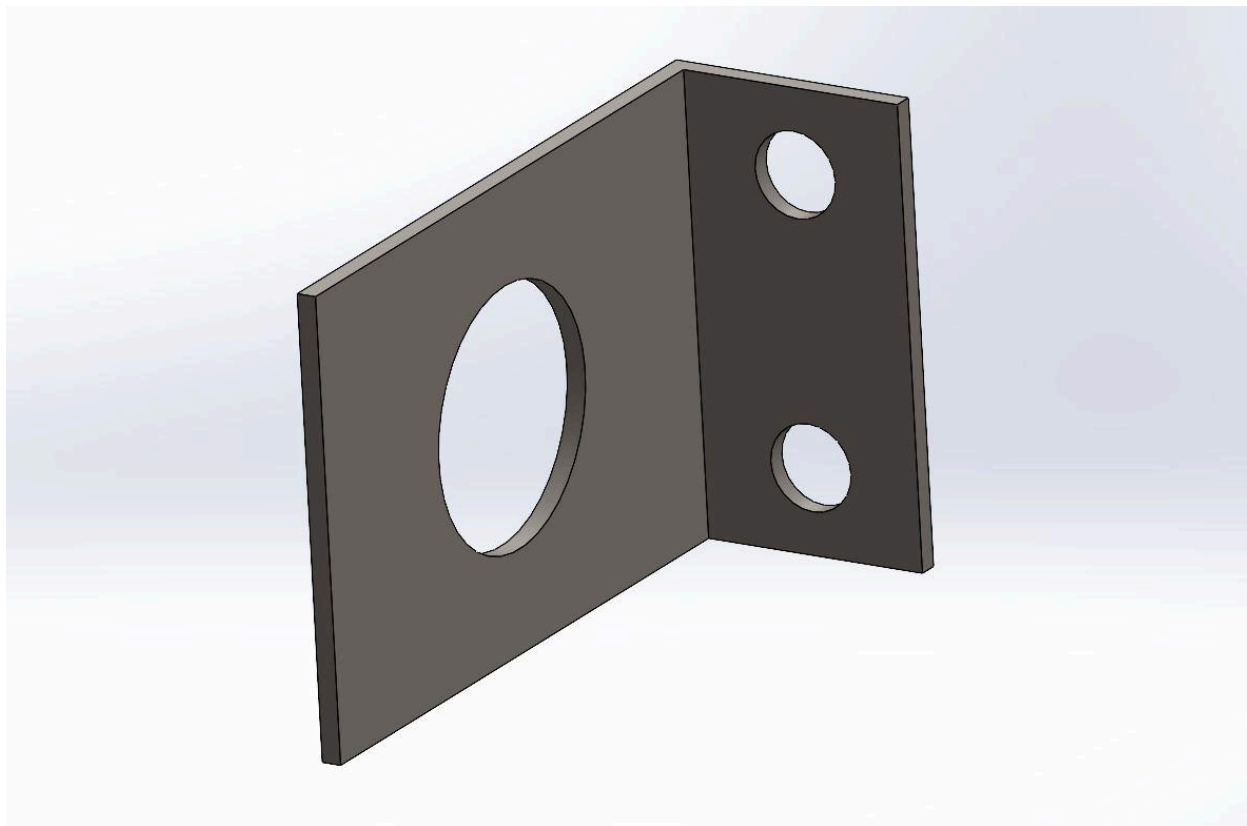


Figure 21 - Shaft Bracket

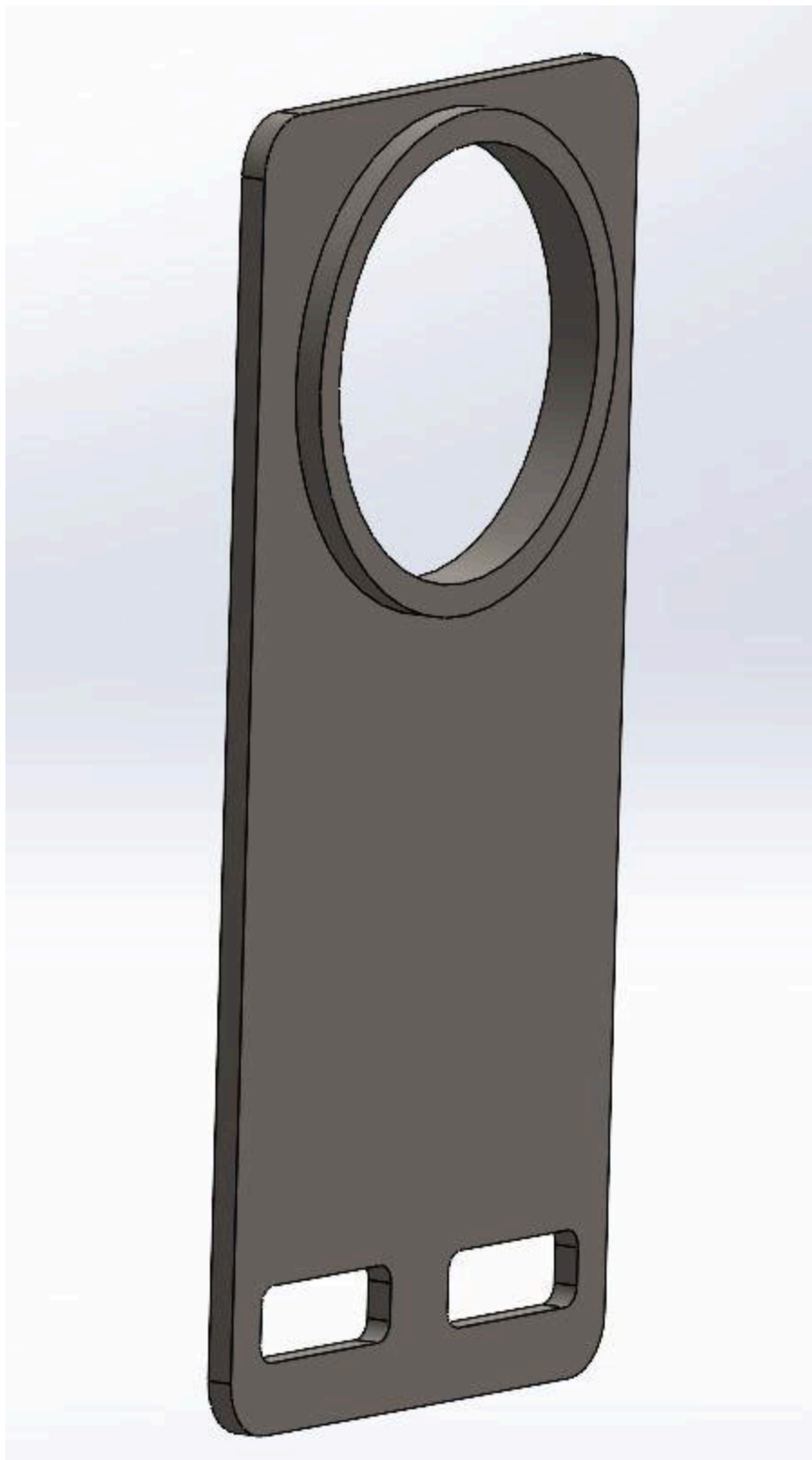


Figure 22 - End Bearing Plate

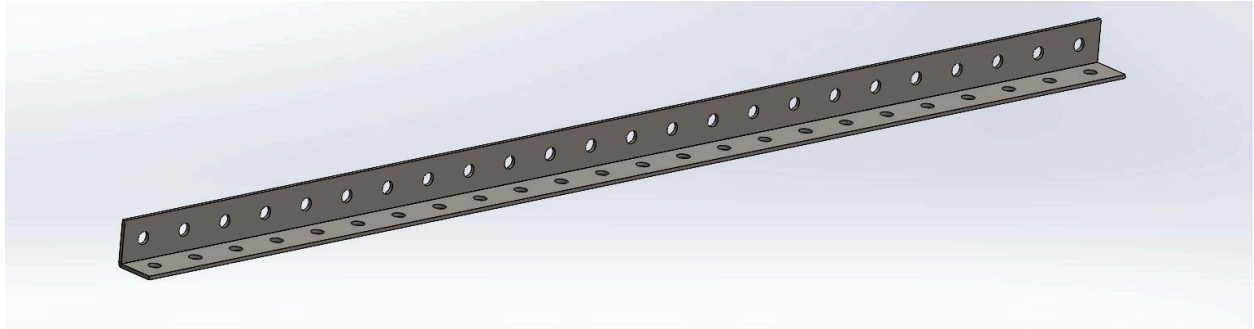


Figure 23 - Angle Track

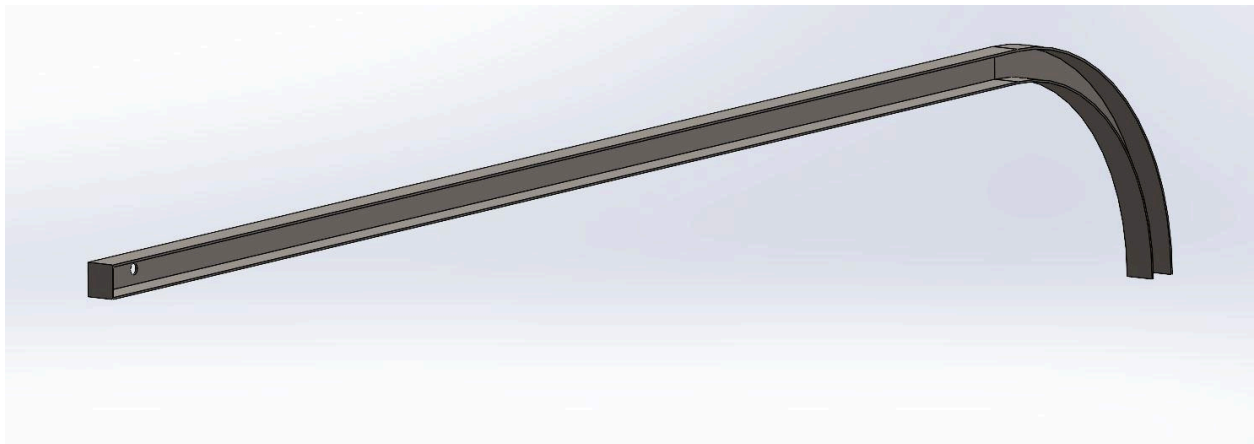


Figure 24 - Horizontal Track

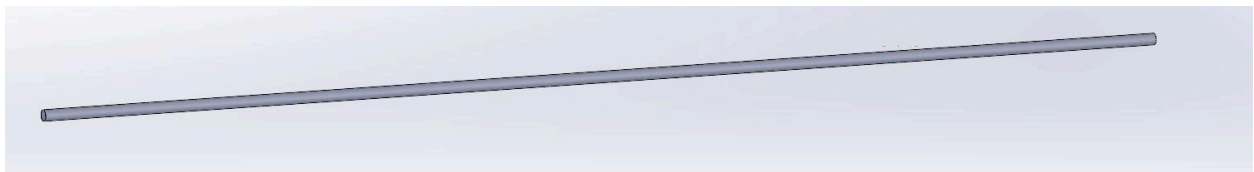


Figure 25 - Shaft

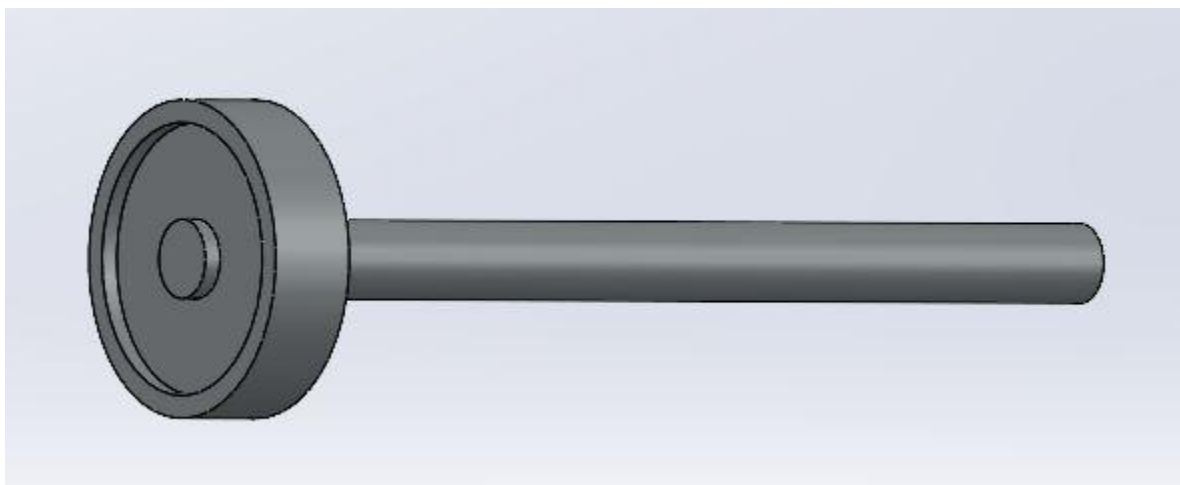


Figure 26 - Roller

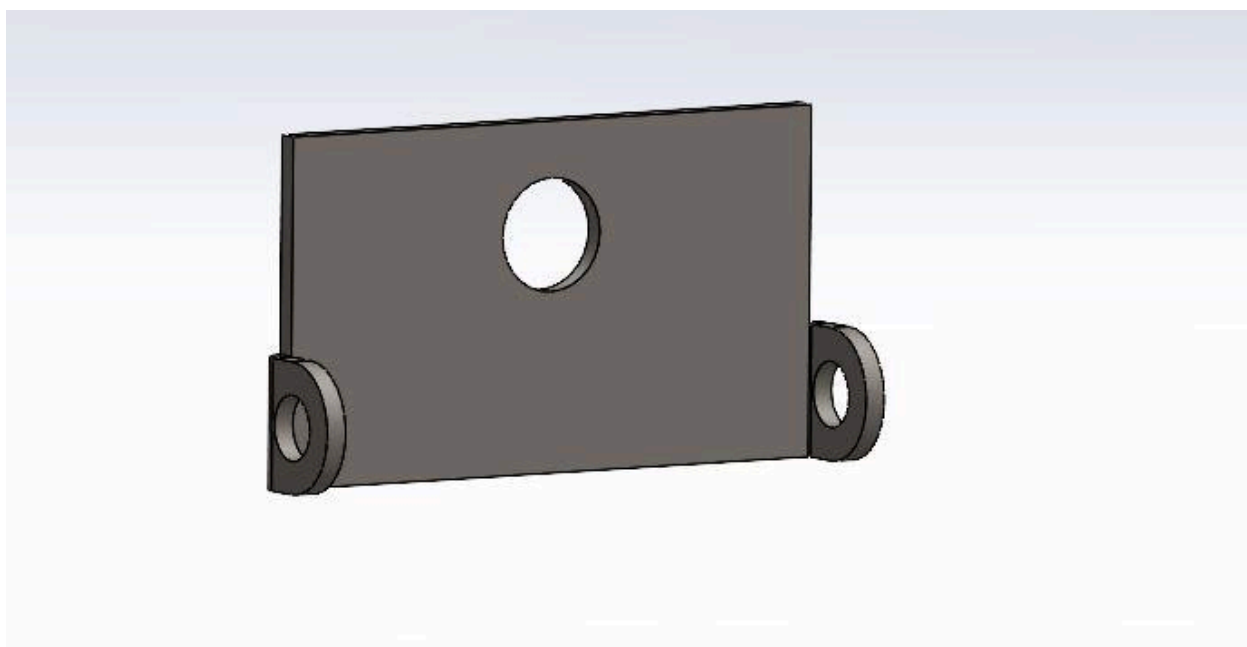


Figure 27 - Bottom Roller Bracket

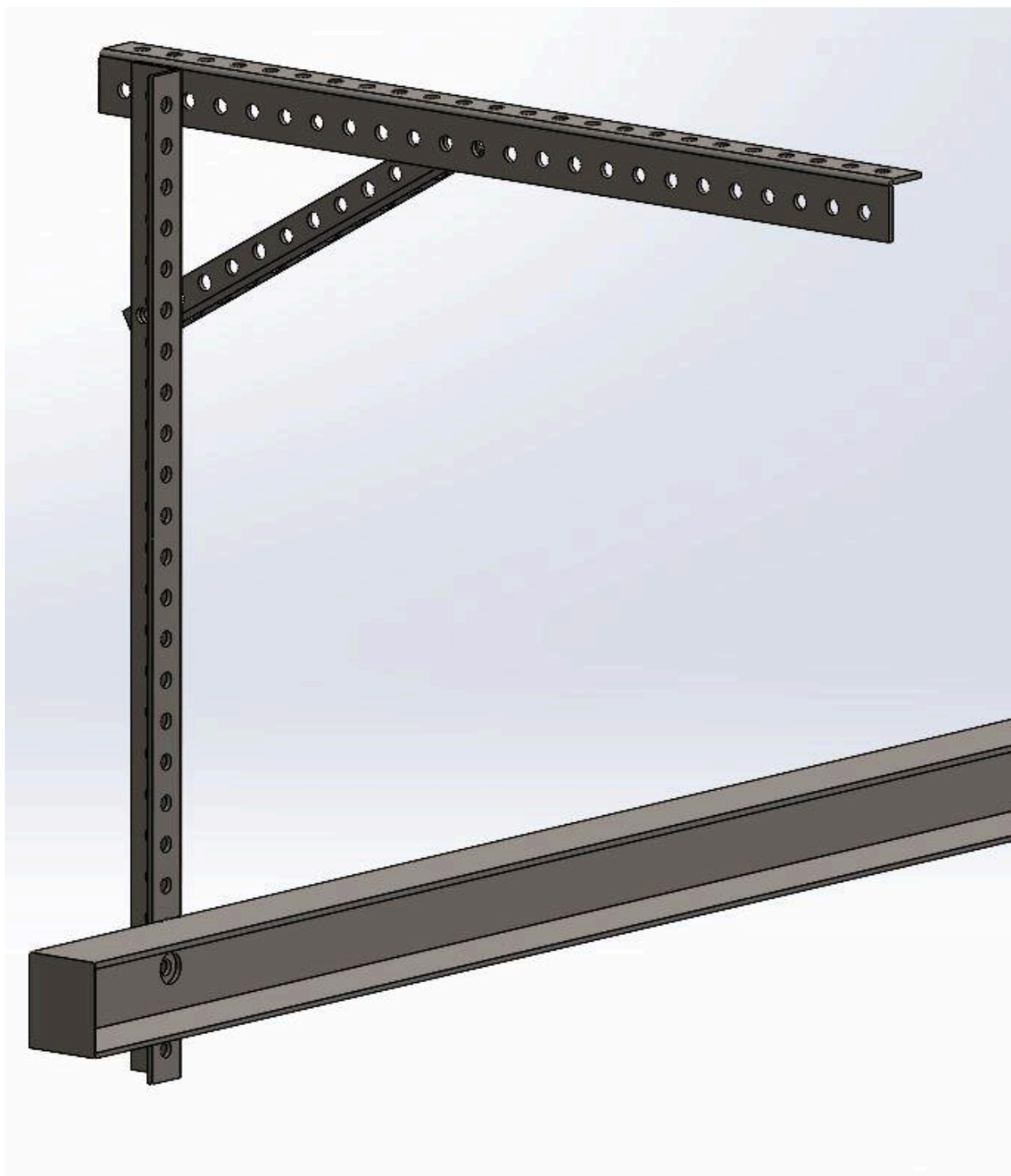


Figure 28 - Rear Track Hanger

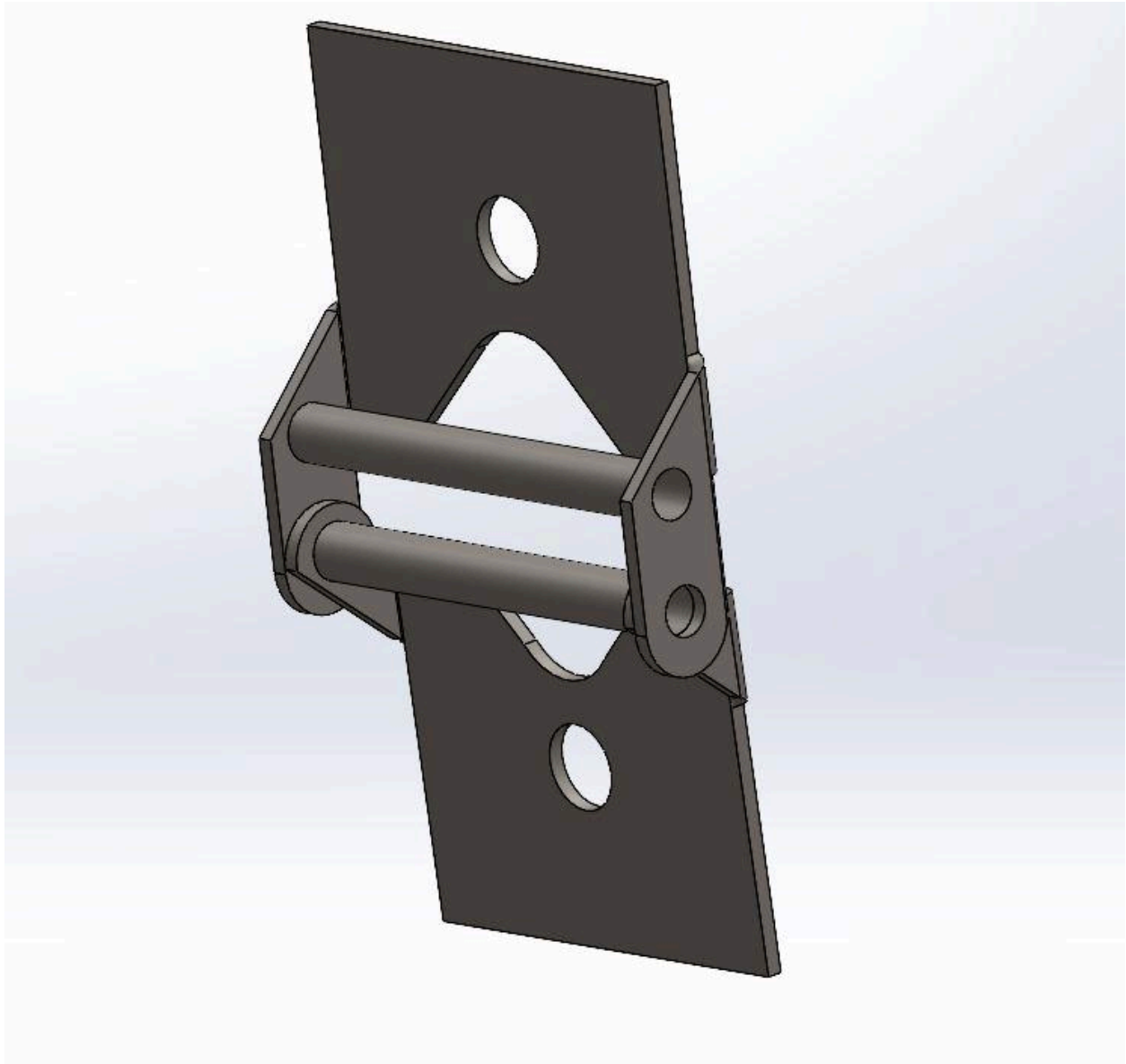


Figure 29 - Hinges

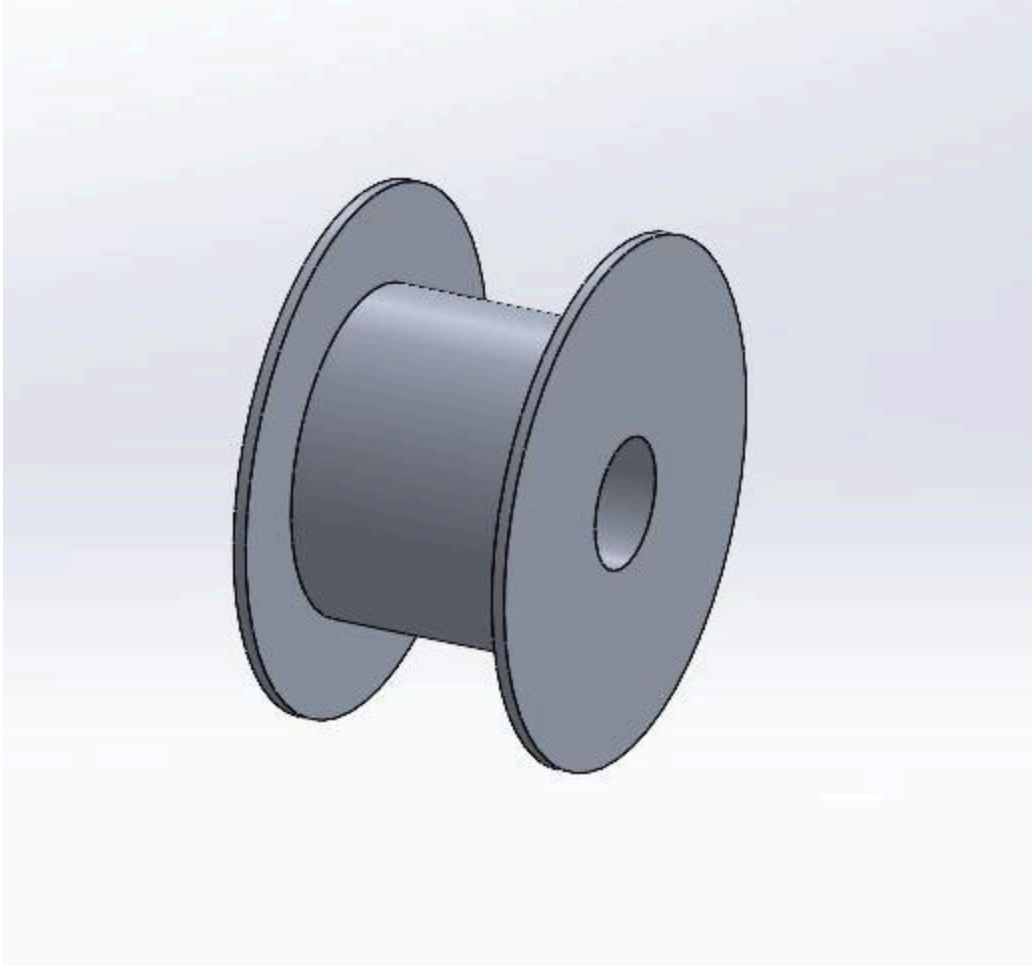


Figure 30 - Cable Drum



Figure 31 - RLS - 8 Bearing

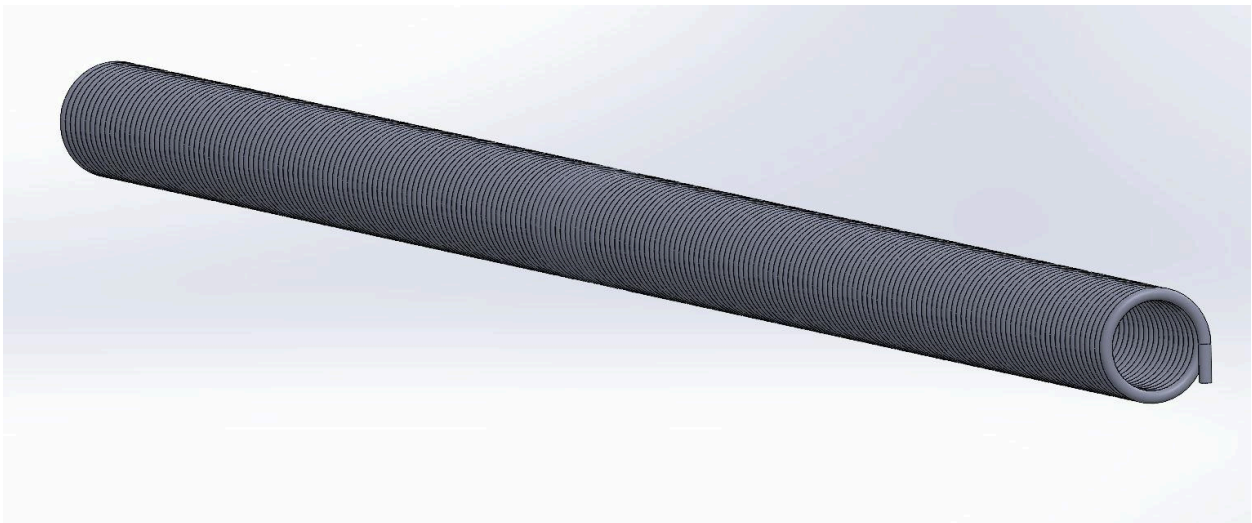


Figure 32 - Torsion Spring

B. Matlab Code

This is the MATLAB code for the vertical track.

```
function check_vertical_track_prompt()

I_input = input('Enter moment of inertia (in^4): ');
b = input('Enter width b (in): ');
h = input('Enter height h (in): ');
t = input('Enter thickness t (in): ');

L = 86; % in
F = 50; % lb
F_bearing = 3.125; % lb
V = 25; % lb
E = 30e6; % psi
d = 0.375; % in (hole diameter)
sigma_allow = 38500; % psi
tau_allow = 38500; % psi

omega = F / L;

% Max Moment
M_max = (omega * L^2) / 12;

% Max Bending Stress
c = h / 2;
sigma_max = (M_max * c) / I_input;

% Shear Calculations
Q = b * t * (0.5) + b * t * (0.870625); % in^3
tau_max = (V * Q) / (I_input * t);

% Bearing Stress
sigma_bearing = F_bearing / (d * t);

% Deflection
ymax = (omega * L^4) / (384 * E * I_input);

% Results
fprintf('--- Vertical Track Analysis ---\n');
fprintf('Max Bending Stress: %.2f psi --> %s\n', sigma_max, pass_fail(sigma_max, sigma_allow));
fprintf('Max Shear Stress: %.2f psi --> %s\n', tau_max, pass_fail(tau_max, tau_allow));
fprintf('Bearing Stress: %.2f psi --> %s\n', sigma_bearing, pass_fail(sigma_bearing, sigma_allow));
fprintf('Max Deflection: %.3f in --> %s\n', ymax, pass_fail(ymax, 0.1));
end
```

```

function result = pass_fail(value, allowable)
    if value <= allowable
        result = 'PASS';
    else
        result = 'FAIL';
    end
end

```

This is the MATLAB code for the horizontal track.

```

function check_horizontal_track_prompt()

    I_input = input('Enter moment of inertia (in^4): ');
    b = input('Enter width b (in): ');
    h = input('Enter height h (in): ');
    t = input('Enter thickness t (in): ');

    L = 117; % in (length of track)
    Lt = 117; % in (used in deflection)
    F = 50; % lb
    F_bearing = 8.33; % lb
    V = 25; % lb
    E = 30e6; % psi
    d = 0.375; % in (hole diameter)
    sigma_allow = 38500; % psi
    tau_allow = 38500; % psi

    omega = F / L;

    % Max Moment
    M_max = (omega * Lt^2) / 12;

    % Max Bending Stress
    c = h / 2;
    sigma_max = (M_max * c) / I_input;

    % Shear Calculations
    Q = b * t * (0.5) + b * t * (0.870625); % in^3
    tau_max = (V * Q) / (I_input * t);

    % Bearing Stress
    sigma_bearing = F_bearing / (d * t);

    % Deflection
    ymax = (omega * Lt^4) / (384 * E * I_input);

    % Results

```

```

fprintf('--- Horizontal Track Analysis ---\n');
fprintf('Max Bending Stress: %.2f psi --> %s\n', sigma_max, pass_fail(sigma_max, sigma_allow));
fprintf('Max Shear Stress: %.2f psi --> %s\n', tau_max, pass_fail(tau_max, tau_allow));
fprintf('Bearing Stress: %.2f psi --> %s\n', sigma_bearing, pass_fail(sigma_bearing, sigma_allow));
fprintf('Max Deflection: %.3f in --> %s\n', ymax, pass_fail(ymax, 0.1));
end

function result = pass_fail(value, allowable)
    if value <= allowable
        result = 'PASS';
    else
        result = 'FAIL';
    end
end
end

```

This is the MATLAB code for the rear track hanger.

```

function check_rear_track_hanger_prompt()
    % Prompt user for inputs
    I = input('Enter moment of inertia I (in^4): ');
    b = input('Enter horizontal leg width b (in): ');
    h = input('Enter vertical leg height h (in): ');
    t = input('Enter thickness t (in): ');

    % Constants
    L = 24; % in (length of hanger)
    F = 150; % lb
    E = 30e6; % psi
    d = 0.375; % in
    pi = 3.1415;
    sigma_allow = 38500; % psi
    tau_allow = 38500; % psi

    % Area and centroid calculations
    A1 = t * h;
    A2 = t * b;
    y1 = t / 2;
    y2 = h + t / 2;

    A_total = A1 + A2;
    y_bar = (A1 * y1 + A2 * y2) / A_total;

    % Outer fiber distance for bending
    c = h/2; % distance to farthest edge

    % Max Bending Moment
    M_max = (F * L) / 8;

```

```

% Bending Stress
sigma = (M_max * c) / I;

% Bearing Stress
sigma_bearing = F / (d * t);

% Shear Stress (approximate for double shear)
tau = F / ((pi/4)*(d^2));

% Deflection
ymax = (F * L^3) / (192 * E * I);

% Output results
fprintf('--- Rear Track Hanger Analysis ---\n');
fprintf('Centroid y?: %.5f in\n', y_bar);
fprintf('Distance to outer fiber c: %.5f in\n', c);
fprintf('Max Bending Stress: %.2f psi --> %s\n', sigma, pass_fail(sigma, sigma_allow));
fprintf('Bearing Stress: %.2f psi --> %s\n', sigma_bearing, pass_fail(sigma_bearing, sigma_allow));
fprintf('Shear Stress: %.2f psi --> %s\n', tau, pass_fail(tau, tau_allow));
fprintf('Max Deflection: %.4f in --> %s\n', ymax, pass_fail(ymax, 0.1));
end

function result = pass_fail(value, allowable)
    if value <= allowable
        result = 'PASS';
    else
        result = 'FAIL';
    end
end
end

```

This is the MATLAB code for the angle track.

```

function check_angle_track_prompt()

    b = input('Enter horizontal leg width b (in): ');
    h = input('Enter vertical leg height h (in): ');
    t = input('Enter thickness t (in): ');
    I = input('Enter moment of inertia I (in^4): ');

    L = 36;      % in (length of track)
    F = 210;     % lb (total distributed load)
    V = 90;     % lb (shear force)
    d = 0.375;   % in (hole diameter)
    E = 30e6;    % psi
    pi = 3.1415; % in
    sigma_allow = 38500; % psi
    tau_allow = 38500; % psi
    c = h/2;

```

```

% Load per unit length
omega = F / L;

% Maximum Moment
M_max = (omega * L^2) / 12;

% Bending Stress
sigma_bending = (M_max * c) / I;

% Bearing Stress
sigma_bearing = V / (d * t);

% Shear Stress
tau_shear = V / ((pi/4)*(d^2));

% Deflection
ymax = (omega * L^4) / (384 * E * I);

% Output results
fprintf('\n--- Angle Track Analysis ---\n');
fprintf(' Max Bending Stress = %8.2f psi --> %s\n', sigma_bending, pass_fail(sigma_bending, sigma_allow));
fprintf(' Bearing Stress    = %8.2f psi --> %s\n', sigma_bearing, pass_fail(sigma_bearing, sigma_allow));
fprintf(' Shear Stress       = %8.2f psi --> %s\n', tau_shear, pass_fail(tau_shear, tau_allow));
fprintf(' Max Deflection     = %8.4f in --> %s\n', ymax, pass_fail(ymax, 0.1));
fprintf('-----\n');
end

function result = pass_fail(value, allowable)
    if value <= allowable
        result = 'PASS';
    else
        result = 'FAIL';
    end
end
end

```

This is the MATLAB code for the spring.

```

% Material and constant values
E = 29.5e6; % psi (converted from MPsi)
ID = 2.0; % in, inside diameter

% Drum specifications
drum_diameters = [4, 5.25, 8]; % inches
torques = [180, 236.23, 360]; % in-lbf
spring_rates = [20.57, 27, 41.14]; % lbf/in

% Wire sizes to evaluate
wire_sizes = [0.187, 0.192, 0.207, 0.218, 0.225, 0.234, 0.243, 0.250, 0.262, 0.273];

```

```

% Loop over each drum
for i = 1:length(drum_diameters)
    fprintf('\n===== Drum Size: %.2f in =====\n', drum_diameters(i));
    T = torques(i); % torque in in-lbf
    K = spring_rates(i); % spring rate in lbf/in

    % Loop over each wire size
    for j = 1:length(wire_sizes)
        d = wire_sizes(j); % wire diameter in inches
        D = d + ID; % mean diameter
        C = D / d; % spring index

        % Step 1: Active coils
        Na = ((d^4)*E)/((10.8)*D*K);
        Nt = Na + 4; % total coils

        % Step 2: Spring length
        spring_length = Nt * d;

        % Step 3: Ki
        Ki = (4*C^2 - C - 1)/(4*C*(C - 1));

        % Step 4: Strength calculations
        Sut = 169 / (d^0.168); % kpsi
        Sy = 0.87 * Sut; % kpsi

        % Step 5: Stress
        stress = (Ki * 32 * T) / (pi * d^3); % in psi (stress)
        stress_kpsi = stress / 1000; % convert to kpsi
        stress_pass = stress_kpsi < Sy;
        fos_stress = Sy / stress_kpsi;

        % Step 6: Tmax
        Tmax = (pi * d^3 * Sy * 1000) / (32 * Ki); % Tmax in in-lbf
        torque_pass = T <= Tmax;
        fos_torque = Tmax / T;

        % Output
        fprintf('\nWire Size: %.3f in\n', d);
        fprintf(' Mean Diameter D: %.3f in\n', D);
        fprintf(' Spring Index C: %.3f\n', C);
        fprintf(' Active Coils Na: %.3f\n', Na);
        fprintf(' Total Coils Nt: %.3f\n', Nt);
        fprintf(' Spring Length: %.3f in\n', spring_length);
        fprintf(' Ki: %.3f\n', Ki);
        fprintf(' Sut: %.2f kpsi, Sy: %.2f kpsi\n', Sut, Sy);
        fprintf(' Stress: %.2f kpsi --> %s (FOS: %.2f)\n', stress_kpsi, ...
            ternary(stress_pass, 'PASS', 'FAIL'), fos_stress);
    end
end

```

```

        fprintf(' Tmax: %.2f in-lbf --> %s (FOS: %.2f)\n', Tmax, ...
            ternary(torque_pass, 'PASS', 'FAIL'), fos_torque);
    end
end

% Ternary helper function
function out = ternary(cond, valTrue, valFalse)
    if cond
        out = valTrue;
    else
        out = valFalse;
    end
end

```